RESTRAINTS

08 SECTION

08-02 ON-BOARD DIAGNOSTIC

FOREWORD

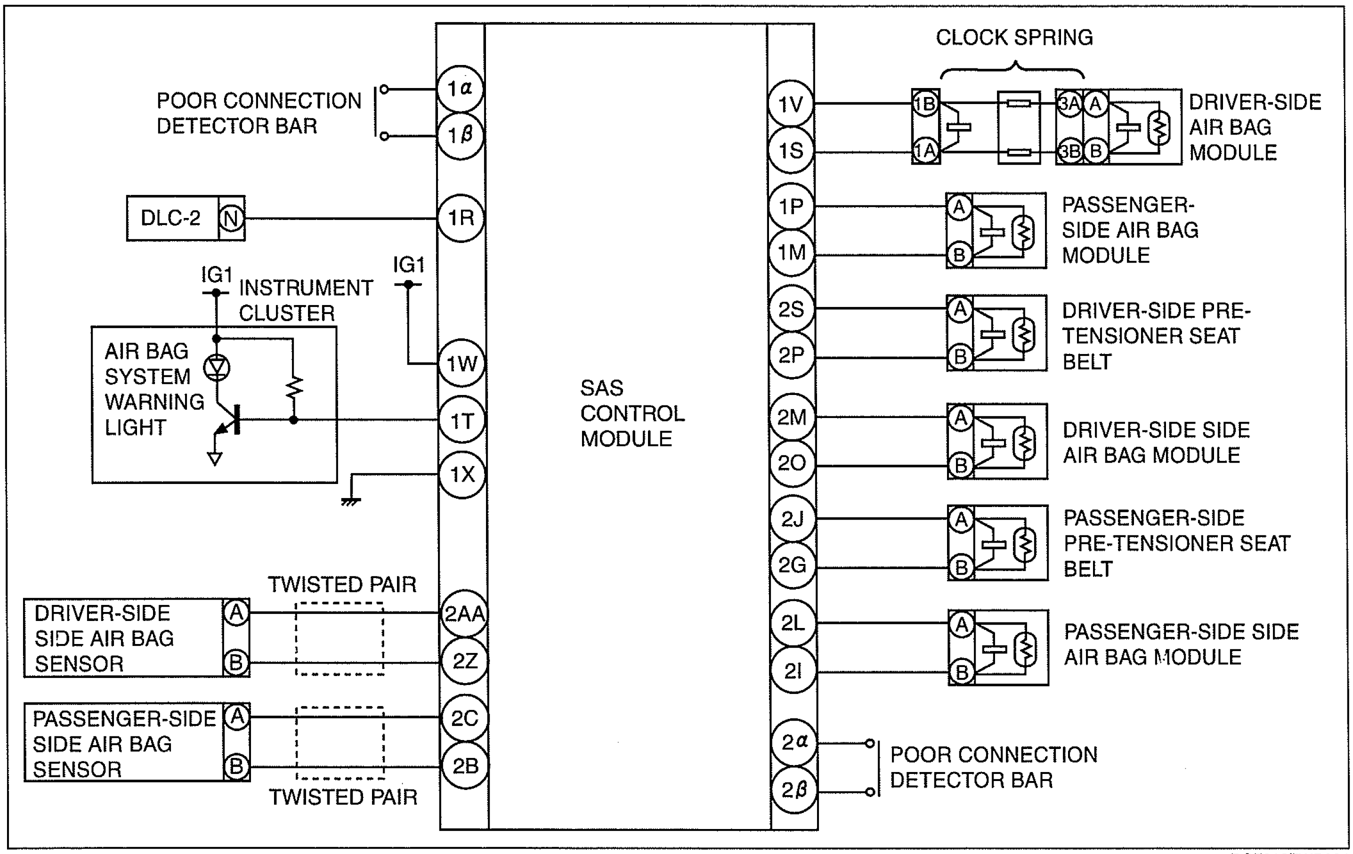
dcf080200000w01

Outline

- The OBD (on-board diagnostic) system has the following functions:
  — Malfunction detection function: Detects malfunctions in the air bag system and outputs DTCs.
  — Data monitor function: Reads out specific input/output signals and the system status.
- Diagnostic DTCs can be read/cleared using the current diagnostic tool.

AIR BAG SYSTEM WIRING DIAGRAM (ON-BOARD DIAGNOSTIC)

dcf080200000w02



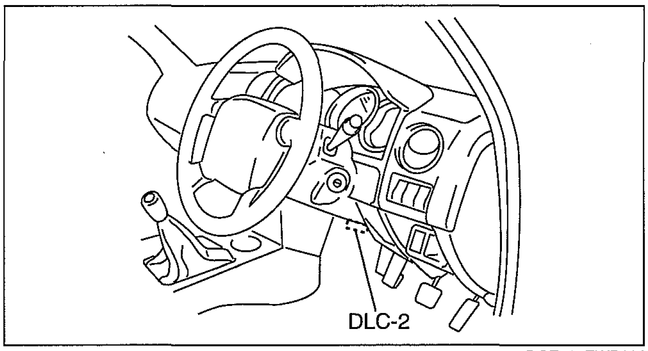
DCF810ZWB001

DTC DISPLAY

dcf080200000w03

Using WDS

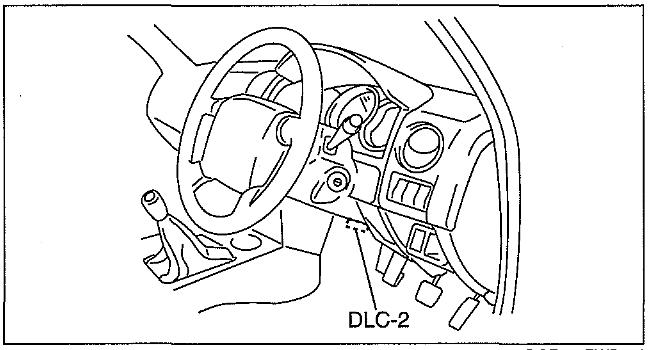
1. Connect the WDS to the DLC-2 connector.
2. Retrieve DTC using the WDS.



DCF802ZWB003

Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "Self Test".
     3. Select "Module".
     4. Select "RCM".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "RCM".
     3. Select "Self Test".



DCF802ZWB003

3. Verify the DTC according to the directions on the screen.
   - If any DTCs are displayed, perform troubleshooting according to the corresponding DTC inspection.
4. After completion of repairs, clear all DTCs stored in the SAS control module. (See 08-02-2 CLEARING DTC.)

CLEARING DTC

dcf080200000w04

Using WDS

1. After repairs have been made, perform the **DTCs reading procedure**.
2. Erase DTC using the WDS.
3. Ensure that the customer's concern has been resolved.

Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "Self Test".
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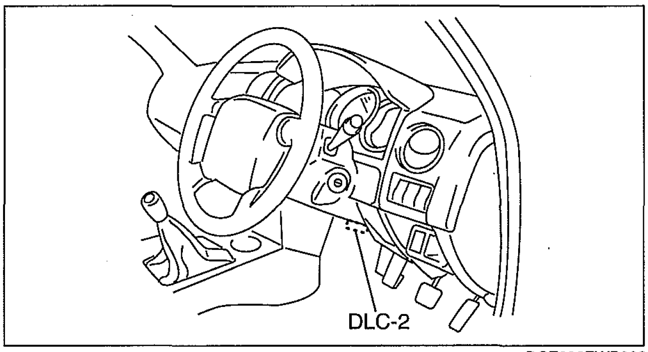
     1. Select "Module Tests".
     2. Select "RCM".
     3. Select "Self Test".



DCF802ZWB003

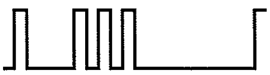
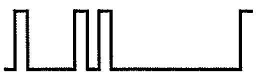
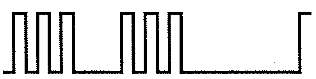
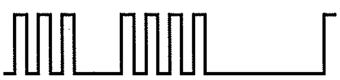
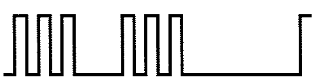
3. Verify the DTC according to the directions on the screen.
4. Press the clear button on the DTC screen to clear the DTC.
5. Verify that no DTCs are displayed.

DTC TABLE

dcf080200000w05

- DTCs are common for present and past malfunction diagnosis.

Note
- When DTCs not shown in the DTC table are displayed, replace the SAS control module.
- If the air bag system warning light does not illuminate or remains illuminated when the engine switch is turned to the ON position, inspect and repair the air bag system warning light circuit, and then confirm that the air bag system warning light is operational.
- The air bag system warning light flashes the DTC pattern for five cycles, and then remains illuminated until the engine switch is turned to the LOCK position.

DTC				System malfunction location	Page
Current diagnostic tool display	Air bag system warning light				
	Flashing pattern		Priority ranking		
B1231	13		3	SAS control module activation (deployment) control freeze	(See 08-02-8 DTC B1231.)
B1317	—	Air bag system warning light is illuminated all the time	—	The SAS control module power supply voltage increases (16.1 V or more)	(See 08-02-8 DTC B1317, B1318.)
B1318	—	Air bag system warning light is illuminated all the time	—	The SAS control module power supply voltage decreases (less than 9 V)	(See 08-02-8 DTC B1317, B1318.)
B1342	12		2	SAS control module	(See 08-02-9 DTC B1342.)
	—	Air bag system warning light is illuminated all the time	1	SAS control module (air bag system warning light DTC 12 is displayed)	
B1869	—	Air bag system warning light is illuminated all the time	1	Air bag system warning light system circuit open	(See 08-02-10 DTC B1869, B1870.)
	—	Air bag system warning light dose not illuminate	—	Air bag system warning light system circuit short to ground	
B1870	—	Air bag system warning light is illuminated all the time	1	Air bag system warning light system circuit short to power supply	
B1877	33		10	Driver-side pre-tensioner seat belt circuit resistance high	(See 08-02-12 DTC B1877, B1878, B1879, B1885.)
B1878				Driver-side pre-tensioner seat belt circuit short to power supply	
B1879				Driver-side pre-tensioner seat belt circuit short to ground	
B1881	34		9	Passenger-side pre-tensioner seat belt circuit resistance high	(See 08-02-14 DTC B1881, B1882, B1883, B1886.)
B1882				Passenger-side pre-tensioner seat belt circuit short to power supply	
B1883				Passenger-side pre-tensioner seat belt circuit short to ground	
B1885	33		10	Driver-side pre-tensioner seat belt circuit resistance low	(See 08-02-12 DTC B1877, B1878, B1879, B1885.)

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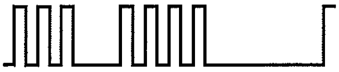
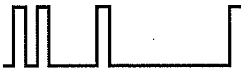
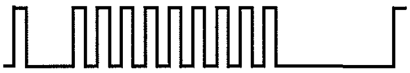
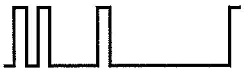
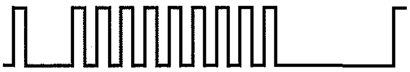
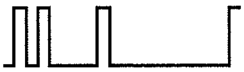
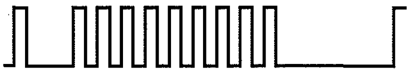
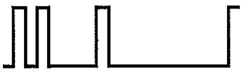
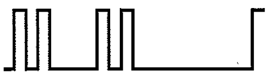
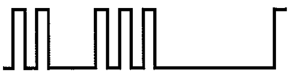
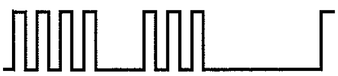
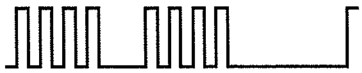
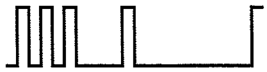
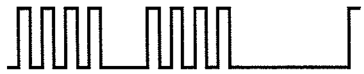
DTC				System malfunction location	Page
Current diagnostic tool display	Air bag system warning light				
	Flashing pattern		Priority ranking		
B1886	34		9	Passenger-side pre-tensioner seat belt circuit resistance low	(See 08-02-14 DTC B1881, B1882, B1883, B1886.)
B1913	19		8	Driver-side air bag module circuit short to body ground	(See 08-02-16 DTC B1913, B1916, B1932, B1934.)
	21		7	Passenger-side air bag module circuit short to body ground	(See 08-02-18 DTC B1913, B1925, B1933, B1935.)
B1916	19		8	Driver-side air bag module circuit short to power supply	(See 08-02-16 DTC B1913, B1916, B1932, B1934.)
B1925	21		7	Passenger-side air bag module circuit short to power supply	(See 08-02-18 DTC B1913, B1925, B1933, B1935.)
B1932	19		8	Driver-side air bag module circuit resistance high	(See 08-02-16 DTC B1913, B1916, B1932, B1934.)
B1933	21		7	Passenger-side air bag module circuit resistance high	(See 08-02-18 DTC B1913, B1925, B1933, B1935.)
B1934	19		8	Driver-side air bag module circuit resistance low	(See 08-02-16 DTC B1913, B1916, B1932, B1934.)
B1935	21		7	Passenger-side air bag module circuit resistance low	(See 08-02-18 DTC B1913, B1925, B1933, B1935.)
B1992	22		12	Driver-side side air bag module circuit short to power supply	(See 08-02-20 DTC B1992, B1993, B1994, B1995.)
B1993				Driver-side side air bag module circuit short to ground	
B1994				Driver-side side air bag module circuit resistance high	
B1995				Driver-side side air bag module circuit resistance low	
B1996	23		11	Passenger-side side air bag module circuit short to power supply	(See 08-02-22 DTC B1996, B1997, B1998, B1999.)
B1997				Passenger-side side air bag module circuit short to ground	
B1998				Passenger-side side air bag module circuit resistance high	
B1999				Passenger-side side air bag module circuit resistance low	
B2444	43		6	Driver-side side air bag sensor system (internal circuit disabled)	(See 08-02-24 DTC B2444, U2017.)

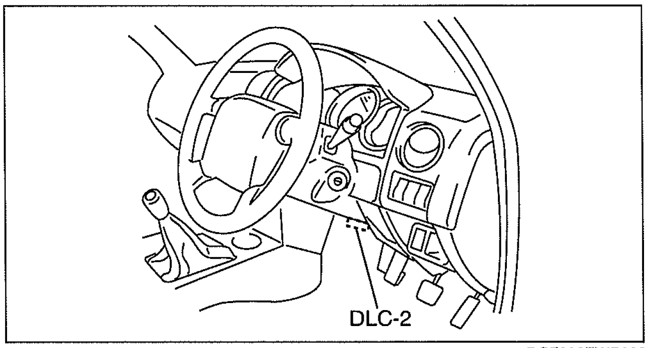
DTC				System malfunction location	Page
Current diagnostic tool display	Air bag system warning light				
	Flashing pattern		Priority ranking		
B2445	44		5	Passenger-side side air bag sensor system (internal circuit disabled)	(See 08-02-25 DTC B2445, U2018.)
B2867	31		4	Poor connection of any SAS control module connectors	(See 08-02-27 DTC B2867.)
U2017	43		6	Driver-side side air bag sensor (communication error)	(See 08-02-24 DTC B2444, U2017.)
U2018	44		5	Passenger-side side air bag sensor (communication error)	(See 08-02-25 DTC B2445, U2018.)

PID/DATA MONITOR DISPLAY

dcf080200000w06

Using WDS

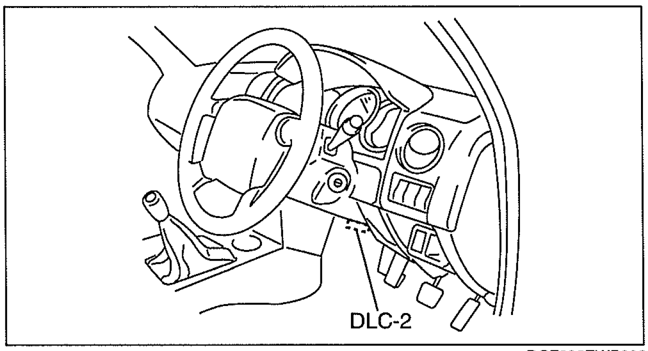
1. Connect the WDS to the DLC-2 connector.
2. Access and monitor PIDs using the WDS.



DCF802ZWB003

Using IDS/PDS

1. Connect the IDS/PDS to the DLC-2 connector.
2. After the vehicle is identified, select the following items from the initialization screen of the IDS/PDS.
   - When using the IDS (notebook PC)
     1. Select the "Toolbox" tab.
     2. Select "DataLogger".
     3. Select "Module".
     4. Select "RCM".
   - When using the PDS (pocket PC)
     1. Select "Module Tests".
     2. Select "RCM".
     3. Select "DataLogger".
3. Select the applicable PID from the PID table.
4. Verify the PID data according to the directions on the screen.



DCF802ZWB003

Note

- The PID data screen function is used for monitoring the calculated value. Therefore, if the monitored value of the output parts is not within the specification, inspection of the monitored value of input parts corresponding to applicable output part control is necessary.In addition, because the system does not display output part malfunction as abnormality in the monitored value, it is necessary to inspect the output part individually.

PID/DATA MONITOR TABLE

dcf080200000w07

PID name (definition)	Unit/condition	Condition/specification	terminal
CONT_RCM (Number of continuous DTC)	—	• DTC is detected: 1—255 • DTC is not detected: 0	—
CRSH_ST_D1 (Driver-side air bag sensor communication state)	OK/ FAULT	• Sensor normal: OK • Sensor communication error: FAULT	2Z, 2AA
CRSH_ST_D2 (Driver-side air bag sensor circuit state)	OK/ FAULT	• Sensor normal: OK • Sensor internal circuit error: FAULT	2Z, 2AA
CRSH_ST_P1 (Passenger-side air bag sensor communication state)	OK/ FAULT	• Sensor normal: OK • Sensor communication error: FAULT	2B, 2C
CRSH_ST_P2 (Passenger-side air bag sensor circuit state)	OK/ FAULT	• Sensor normal: OK • Sensor internal circuit error: FAULT	2B, 2C
D_PTENSFLT (Driver-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	• Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness circuit resistance high: OPEN • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness normal: Normal	2P, 2S
DABAGR (Driver-side air bag module resistance)	ohm	Under any condition: 1.5—3.7 ohms	1S, 1V
DR_PTENS (Driver-side pre-tensioner seat belt resistance)	ohm	Under any condition: 1.5—3.1 ohms	2P, 2S
DS_AB (Driver-side side air bag module resistance)	ohm	Under any condition: 1.4—3.2 ohms	2M, 2O
DS_AB_ST (Driver-side side air bag module circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	• Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	2M, 2O
DS1_STAT (Driver-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	• Related wiring harness short to ground: SHRT GND • Related wiring harness short to power supply: SHRT B+ • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	1S, 1V
DSB_P_ST (On demand driver-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	• Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness circuit resistance high: OPEN • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness normal: Normal	2P, 2S
IG_V_2 (System IG1 voltage value)	V	Engine switch to ON position: B+	1W
OD_CRST_D1 (On demand driver-side side air bag sensor communication state)	OK/ FAULT	• Sensor normal: OK • Sensor communication error: FAULT	2Z, 2AA
OD_CRST_D2 (On demand driver-side side air bag sensor circuit state)	OK/ FAULT	• Sensor normal: OK • Sensor internal circuit error: FAULT	2Z, 2AA
OD_CRST_P1 (On demand passenger-side side air bag sensor communication state)	OK/ FAULT	• Sensor normal: OK • Sensor communication error: FAULT	2B, 2C
OD_CRST_P2 (On demand passenger-side side air bag sensor circuit state)	OK/ FAULT	• Sensor normal: OK • Sensor internal circuit error: FAULT	2B, 2C

PID name (definition)	Unit/condition	Condition/specification	terminal
OD_DAB1_ST (On demand driver-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	• Related wiring harness short to ground: SHRT_GND • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	1S, 1V
OD_DSAB_ST (On demand driver-side side air bag circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	• Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	2M, 2O
OD_PAB1_ST (On demand passenger-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	• Related wiring harness short to ground: SHRT_GND • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	1M, 1P
OD_PSAB_ST (On demand passenger-side side air bag sensor circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	• Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	2I, 2L
P_PTENSFLT (Passenger-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	• Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness circuit resistance high: OPEN • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness normal: Normal	2G, 2J
PABAGR (Passenger-side air bag module resistance)	ohm	Under any condition: 1.4—2.9 ohms	1M, 1P
PS_AB (Passenger-side side air bag module resistance)	ohm	Under any condition: 1.4—3.2 ohms	2I, 2L
PS_AB_ST (Passenger-side side air bag sensor circuit state)	SHRT_B+ SHRT_GND OPEN SQ_LOWRES Normal	• Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	2I, 2L
PS_PTENS (Passenger-side pre-tensioner seat belt resistance)	ohm	Under any condition: 1.5—3.1 ohms	2G, 2J
PS1_STAT (Passenger-side air bag module circuit state)	SHRT_GND SHRT_B+ OPEN SQ_LOWRES Normal	• Related wiring harness short to ground: SHRT_GND • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness circuit resistance high: OPEN • Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness normal: Normal	1M, 1P
PSB_P_ST (On demand passenger-side pre-tensioner seat belt circuit state)	SQ_LOWRES OPEN SHRT_B+ SHRT_GND Normal	• Air bag module circuit resistance low: SQ_LOWRES • Related wiring harness circuit resistance high: OPEN • Related wiring harness short to power supply: SHRT_B+ • Related wiring harness short to ground: SHRT_GND • Related wiring harness normal: Normal	2G, 2J

DTC B1231

dcf080200000w08

DTC B1231	SAS control module activation (deployment) control freeze
DETECTION CONDITION	**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • SAS control module determined collision

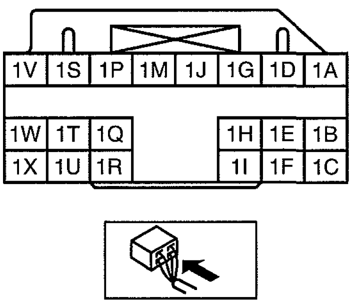
Diagnostic procedure

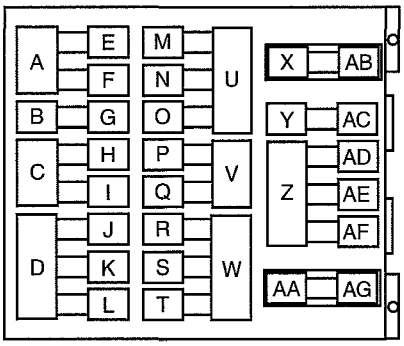
ACTION
Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)

DTC B1317, B1318

dcf080200000w09

DTC	B1317	SAS control module power supply voltage increases (16.1 V or more)
DTC	B1318	SAS control module power supply voltage decreases (less than 9 V)
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • When the SAS control module power supply voltage is not **within 9—16 V**.
POSSIBLE CAUSE		• Open or short circuit in wiring harness between battery and SAS control module • ENGINE 15 A fuse malfunction • Battery malfunction • SAS control module malfunction





Diagnostic procedure

Step	Inspection		Action
1	**INSPECT FUSE** • Remove the ENGINE 15 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	Replace the fuse.
2	**INSPECT BATTERY** • Measure the battery positive voltage. • Is the voltage **9 V—16 V**?	Yes	Go to the next step.
		No	The battery has a malfunction. Inspect the charge/discharge system.
3	**INSPECT WIRING HARNESS BETWEEN BATTERY AND FUSE BLOCK** • Turn the engine switch to the ON position. • Measure the fuse block terminal W voltage. • Is the voltage **9 V—16 V**?	Yes	Install the fuse, then go to the next step.
		No	Repair the wiring harness between the fuse block and engine switch.

Step	Inspection		Action
4	**INSPECT FUSE BLOCK** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. (Vehicles with passenger-side air bag) • Disconnect the passenger-side air bag module connector. (Vehicles with passenger-side air bag) • Disconnect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Disconnect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Remove the console. • Disconnect the all SAS control module connectors. • Turn the engine switch to the ON position. • Measure the SAS control module terminal 1W voltage. • Is the voltage **9 V—16 V**?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Repair the wiring harness between the fuse block and SAS control module.

DTC B1342

dcf080200000w10

DTC B1342	SAS control module
DETECTION CONDITION	**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Malfunction in the SAS control module internal circuit
POSSIBLE CAUSE	• SAS control module malfunction

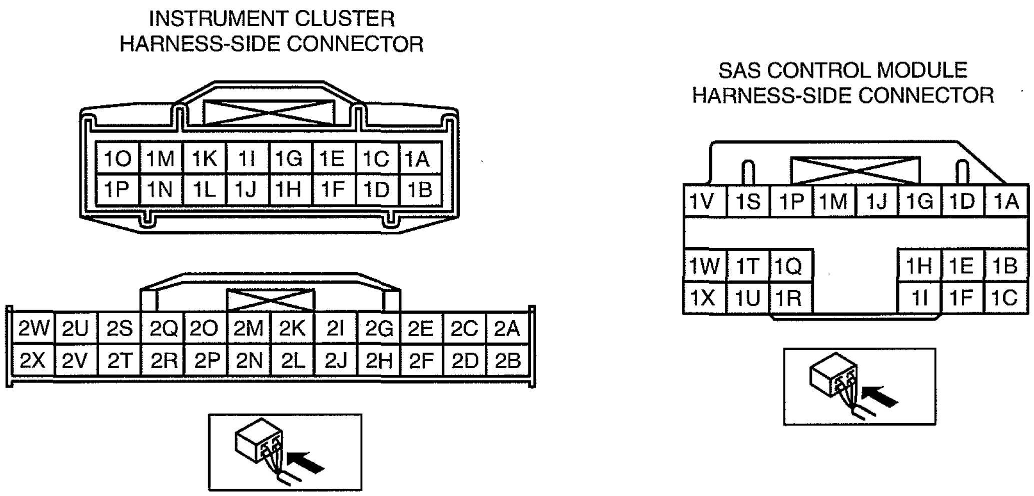
Diagnostic procedure

ACTION
Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)

DTC B1869, B1870

dcf080200000w11

DTC	B1869	**Air bag system warning light system circuit open or short to ground**
	B1870	**Air bag system warning light system circuit short to power supply**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Malfunction in air bag system warning light circuit
POSSIBLE CAUSE		• METER 15 A fuse malfunction • Instrument cluster malfunction • Malfunction of connectors between instrument cluster and SAS control module • Open or short circuit in wiring harness between METER 15 A fuse and instrument cluster • Open or short circuit in wiring harness between instrument cluster and SAS control module • SAS control module malfunction



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT METER 15 A FUSE** • Turn engine switch to LOCK position. • Disconnect the negative battery cable. • Remove the METER 15 A fuse. • Is fuse normal?	Yes	Reinstall the METER 15 A fuse, then go to the next step.
		No	Replace the METER 15 A fuse.
2	**INSPECT FOR CONTINUITY BETWEEN METER 15 A FUSE AND INSTRUMENT CLUSTER** • Connect the negative battery cable. • Turn engine switch to ON position. • Measure voltage at instrument cluster connector terminal 2C. • Is voltage **9 V or more**?	Yes	Go to the next step.
		No	Repair the wiring harness.

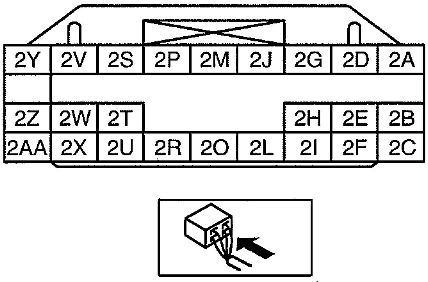
STEP	INSPECTION		ACTION
3	**INSPECT WIRING HARNESS BETWEEN INSTRUMENT CLUSTER AND SAS CONTROL MODULE** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. (Vehicles with passenger-side air bag) • Disconnect the passenger-side air bag module connector. (Vehicles with passenger-side air bag) • Disconnect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Disconnect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Remove the console. • Disconnect the all SAS control module connectors. • Disconnect the instrument cluster. • Inspect following wiring harness between SAS control module and instrument cluster terminals for short to ground, short to power supply, and open circuit: — 1T—1H • Is wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
4	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect instrument cluster. • Turn the engine switch to ON position. • Is air bag system warning light illuminated?	Yes	Go to the next step.
		No	Replace the instrument cluster. (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
5	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Using a jumper wire, cause a short circuit between instrument cluster terminal 1H and ground. • Does air bag system warning light go out?	Yes	Go to the next step.
		No	Replace the instrument cluster. (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
6	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Connect the all SAS control module connectors. • Connect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Connect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Connect the passenger-side air bag module connector. (Vehicles with passenger-side air bag) • Connect the clock spring connector. • Turn the engine switch to ON position. • Verify that the air bag system warning light turns off **approx. 2 s** after illuminating for **approx. 6 s**. • Is the result normal?	Yes	The system is normal at present. (Clear the malfunction from the memory.)
		No	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)

DTC B1877, B1878, B1879, B1885

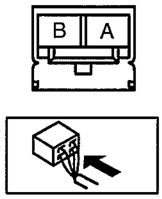
dcf080200000w12

DTC	B1877	**Driver-side pre-tensioner seat belt system resistance high**
	B1878	**Driver-side pre-tensioner seat belt system circuit short to power supply**
	B1879	**Driver-side pre-tensioner seat belt system circuit short to ground**
	B1885	**Driver-side pre-tensioner seat belt system resistance low**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Resistance other than **1.5—3.1 ohms** detected in driver-side pre-tensioner seat belt circuit • Malfunction in the wiring harness between driver-side pre-tensioner seat belt and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between driver-side pre-tensioner seat belt and SAS control module • Driver-side pre-tensioner seat belt malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



DRIVER-SIDE PRE-TENSIONER SEAT BELT WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT DRIVER-SIDE PRE-TENSIONER SEAT BELT** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — DR_PTENS • Is the resistance of the driver-side pre-tensioner seat belt normal? — Resistance: **1.5—3.1 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT DRIVER-SIDE PRE-TENSIONER SEAT BELT CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the driver-side pre-tensioner seat belt connector. • Is there any malfunction of the driver-side pre-tensioner seat belt connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

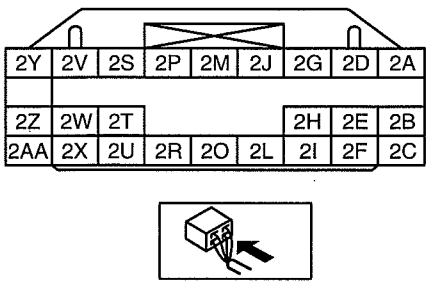
<table>
<tr><th>STEP</th><th>INSPECTION</th><th colspan="2">ACTION</th></tr>
<tr><td rowspan="2">3</td><td rowspan="2">VERIFY WHETHER MALFUNCTION IS IN DRIVER-SIDE PRE-TENSIONER SEAT BELT OR RELATED WIRING HARNESS
• Connect leads of SST (Fuel and thermometer checker) or apply 2-ohm resistor to driver-side pre-tensioner seat belt connector terminal A and B.
• Set resistance of SST (Fuel and thermometer checker) to the 2-ohm position.
• Connect the negative battery cable.
• Turn the engine switch to ON position.
• Are DTCs B1877, B1878, B1879, and/or B1885 indicated?</td><td>Yes</td><td>Go to the next step.</td></tr>
<tr><td>No</td><td>Replace the driver-side pre-tensioner seat belt.
(See 08-10-5 SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)</td></tr>
<tr><td rowspan="2">4</td><td rowspan="2">INSPECT WIRING HARNESS BETWEEN DRIVER-SIDE PRE-TENSIONER SEAT BELT MODULE AND SAS CONTROL MODULE
• Turn the engine switch to the LOCK position.
• Disconnect the negative battery cable and wait for 1 min or more.
• Disconnect the driver-side pre-tensioner seat belt connector.
• Remove the console.
• Disconnect the SAS control module connectors.
• Inspect the wiring harness between SAS control module terminal 2S and driver-side pre-tensioner seat belt terminal A, SAS control module terminal 2P and driver-side pre-tensioner seat belt terminal B for the following:
— Short to ground
— Short to power supply
— Open circuit
• Is the wiring harness normal?</td><td>Yes</td><td>Go to the next step.</td></tr>
<tr><td>No</td><td>Replace the air bag wiring harness.</td></tr>
<tr><td rowspan="2">5</td><td rowspan="2">INSPECT AIR BAG SYSTEM WARNING LIGHT
• Connect the all SAS control module connectors.
• Connect the driver and passenger-side pre-tensioner seat belt connectors.
• Connect the negative battery cable.
• Turn the engine switch to ON position.
• Are DTCs B1877, B1878, B1879, and/or B1885 indicated?</td><td>Yes</td><td>Replace the SAS control module.
(See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)</td></tr>
<tr><td>No</td><td>The system is normal at present. (Clear the malfunction from the memory.)</td></tr>
</table>

DTC B1881, B1882, B1883, B1886

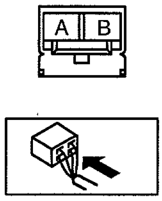
dcf080200000w13

DTC	B1881	Passenger-side pre-tensioner seat belt circuit resistance high
	B1882	Passenger-side pre-tensioner seat belt circuit short to power supply
	B1883	Passenger-side pre-tensioner seat belt circuit short to body ground
	B1886	Passenger-side pre-tensioner seat belt circuit resistance low
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Resistance other than **1.5—3.1 ohms** detected in passenger-side pre-tensioner seat belt circuit • Malfunction in wiring harness between passenger-side pre-tensioner seat belt and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between passenger-side pre-tensioner seat belt and SAS control module • Passenger-side pre-tensioner seat belt malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



PASSENGER-SIDE PRE-TENSIONER SEAT BELT WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT PASSENGER-SIDE PRE-TENSIONER SEAT BELT** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — PS_PTENS • Is the resistance of the passenger-side pre-tensioner seat belt normal? — Resistance: **1.5—3.1 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT PASSENGER-SIDE PRE-TENSIONER SEAT BELT CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the passenger-side pre-tensioner seat belt connector. • Is there any malfunction of the passenger-side pre-tensioner seat belt connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

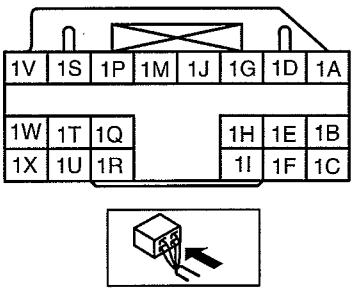
STEP	INSPECTION		ACTION
3	**VERIFY WHETHER MALFUNCTION IS IN PASSENGER-SIDE PRE-TENSIONER SEAT BELT OR RELATED WIRING HARNESS** • Connect the leads of the **SST** (Fuel and thermometer checker) or apply **2-ohm** resistance to passenger-side pre-tensioner seat belt connector terminals A and B. • Set the resistance of the **SST** (Fuel and thermometer checker) to the **2-ohm** position. • Connect the negative battery cable. • Turn the engine switch to the ON position. • Are DTCs B1881, B1882, B1883, and/or B1886 indicated?	Yes	Go to the next step.
		No	Replace the passenger-side pre-tensioner seat belt. (See 08-10-5 SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
4	**INSPECT WIRING HARNESS BETWEEN PASSENGER-SIDE PRE-TENSIONER SEAT BELT MODULE AND SAS CONTROL MODULE** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the driver-side pre-tensioner seat connector. • Remove the console. • Disconnect the SAS control module connectors. • Inspect the wiring harness between SAS control module terminal 2J and passenger-side pre-tensioner seat belt terminal A, SAS control module terminal 2G and passenger-side pre-tensioner seat belt terminal B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
5	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect the all SAS control module connectors. • Connect the driver and passenger-side pre-tensioner seat belt connectors. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1881, B1882, B1883, and/or B1886 indicated?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	The system is normal at present. (Clear the malfunction from the memory.)

DTC B1913, B1916, B1932, B1934

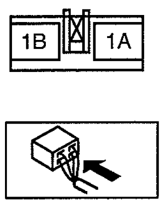
dcf080200000w14

DTC	B1913	**Driver-side air bag module system circuit short to ground**
	B1916	**Driver-side air bag module system circuit short to power supply**
	B1932	**Driver-side air bag module system resistance high**
	B1934	**Driver-side air bag module system resistance low**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • resistance other than **1.5—3.7 ohms** detected in driver-side air bag module circuit • Malfunction in wiring harness between driver-side air bag module and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between clock spring and SAS control module • Clock spring malfunction • Driver-side air bag module malfunction • SAS control module malfunction

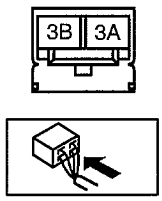
SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



CLOCK SPRING WIRING HARNESS-SIDE CONNECTOR



DRIVER-SIDE AIR BAG MODULE WIRING HARNESS-SIDE CONNECTOR (CLOCK SPRING)



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT DRIVER-SIDE AIR BAG MODULE** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — DABAGR • Is the resistance of the driver-side air bag module normal? — Resistance: **1.5—3.7 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT DRIVER-SIDE AIR BAG MODULE CONNECTOR (CLOCK SPRING)** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the driver-side air bag module. • Is there any malfunction of the driver-side air bag module connector?	Yes	Replace the air bag wiring harness and/or clock spring.
		No	Go to the next step.

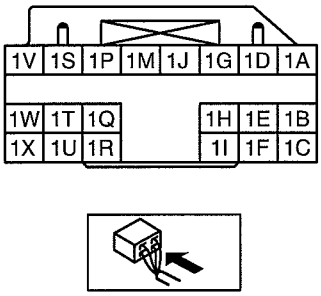
STEP	INSPECTION		ACTION
3	**VERIFY WHETHER MALFUNCTION IS IN DRIVER-SIDE AIR BAG MODULE OR RELATED WIRING HARNESS** • Connect the leads of **SST** (Fuel and thermometer checker) or apply **2-ohm** resistance to driver-side air bag module connector terminals 3A and 3B. • Set the resistance of **SST** (Fuel and thermometer checker) to **2-ohm** position. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1913, B1916, B1932 and/or B1934?	Yes	Go to the next step.
		No	Replace the driver-side air bag module. (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION.)
4	**INSPECT CLOCK SPRING** • Inspect the clock spring. (See 08-10-8 CLOCK SPRING INSPECTION.) • Is the clock spring normal?	Yes	Go to the next step.
		No	Replace the clock spring. (See 08-10-7 CLOCK SPRING REMOVAL/INSTALLATION.)
5	**INSPECT WIRING HARNESS BETWEEN CLOCK SPRING AND SAS CONTROL MODULE** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. • Disconnect the passenger-side air bag module connector. (Vehicles with passenger-side air bag) • Disconnect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Disconnect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Remove the console. • Disconnect the all SAS control module connectors. • Inspect the wiring harness between SAS control module terminal 1S and clock spring terminal 1A, SAS control module terminal 1V and clock spring terminal 1B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
6	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect the all SAS control module connectors. • Connect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Connect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Connect the passenger-side air bag module connector. (Vehicles with passenger-side air bag) • Connect the clock spring connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1913, B1916, B1932 and/or B1934?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)
		No	The system is normal at present. (Clear the malfunction from the memory.)

DTC B1913, B1925, B1933, B1935

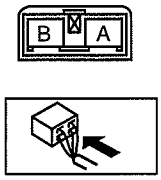
dcf080200000w15

DTC	B1913	**Passenger-side air bag module circuit short to ground**
	B1925	**Passenger-side air bag module circuit short to power supply**
	B1933	**Passenger-side air bag module circuit resistance high**
	B1935	**Passenger-side air bag module circuit resistance low**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Resistance other than **1.4—2.9 ohms** detected in passenger-side air bag module circuit • Malfunction in wiring harness between passenger-side air bag module and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between passenger-side air bag module and SAS control module • Passenger-side air bag module malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



PASSENGER-SIDE AIR BAG MODULE WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT PASSENGER-SIDE AIR BAG MODULE** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — PABAGR • Is the resistance of the passenger-side air bag module normal? — Resistance: **1.4—2.9 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT PASSENGER-SIDE AIR BAG MODULE CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more.** • Remove the glove compartment. • Disconnect the passenger-side air bag module connector. • Is there any malfunction of the passenger-side air bag module connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

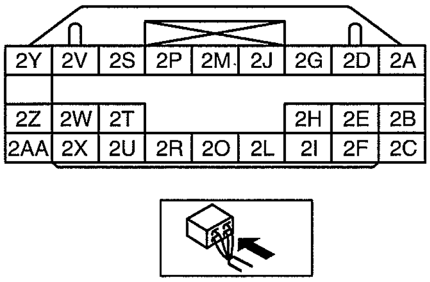
STEP	INSPECTION		ACTION
3	**VERIFY WHETHER MALFUNCTION IS IN PASSENGER-SIDE AIR BAG MODULE OR RELATED WIRING HARNESS** • Connect the leads of the **SST** (Fuel and thermometer checker) or apply **2-ohm** resistance to passenger-side air bag module connector terminals A and B. • Set the resistance of the **SST** (Fuel and thermometer checker) to the **2-ohm** position. • Connect the negative battery cable. • Turn the engine switch to the ON position. • Are DTCs B1913, B1925, B1933 and/or B1935 indicated?	Yes	Go to the next step.
		No	Replace the passenger-side air bag module. (See 08-10-5 PASSENGER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION.)
4	**INSPECT WIRING HARNESS BETWEEN PASSENGER-SIDE AIR BAG MODULE AND SAS CONTROL MODULE** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the passenger-side air bag module connector. • Remove the console. • Disconnect the SAS control module connectors. • Inspect the wiring harness between SAS control module terminal 1P and passenger-side air bag module terminal A, SAS control module terminal 1M and passenger-side air bag module terminal B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
5	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect the all SAS control module connectors. • Connect the passenger-side air bag module connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1913, B1925, B1933 and/or B1935 indicated?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)
		No	The system is normal at present. (Clear the malfunction from the memory.)

DTC B1992, B1993, B1994, B1995

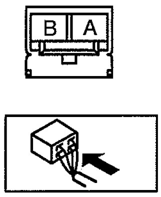
dcf080200000w16

DTC	B1992	**Driver-side side air bag module circuit short to power supply**
	B1993	**Driver-side side air bag module circuit short to ground**
	B1994	**Driver-side side air bag module circuit resistance high**
	B1995	**Driver-side side air bag module circuit resistance low**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Resistance other than **1.4—3.2 ohms** detected in driver-side side air bag module circuit • Malfunction in wiring harness between driver-side side air bag module and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between driver-side side air bag module and SAS control module • Driver-side side air bag module malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



DRIVER-SIDE SIDE AIR BAG MODULE WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT DRIVER-SIDE SIDE AIR BAG MODULE** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — DS_AB • Is the resistance of the driver-side side air bag module normal? — Resistance: **1.4—3.2 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT DRIVER-SIDE SIDE AIR BAG MODULE CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the driver-side side air bag module. • Disconnect the driver-side side air bag module connector. • Is there any malfunction of the driver-side side air bag module connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY WHETHER MALFUNCTION IS IN DRIVER-SIDE SIDE AIR BAG MODULE OR RELATED WIRING HARNESS** • Connect the leads of the **SST** (Fuel and thermometer checker) or apply **2-ohm** resistance to driver-side side air bag module connector terminals A and B. • Set the resistance of the **SST** (Fuel and thermometer checker) to the **2-ohm** position. • Connect the negative battery cable. • Turn the engine switch to the ON position. • Are DTCs B1992, B1993, B1994, and/or B1995 indicated?	Yes	Go to the next step.
		No	Replace the driver-side side air bag module. (See 08-10-5 SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
4	**INSPECT WIRING HARNESS BETWEEN DRIVER-SIDE SIDE AIR BAG MODULE AND SAS CONTROL MODULE** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the driver-side side air bag module connector. • Remove the console. • Disconnect the SAS control module connectors. • Inspect the wiring harness between SAS control module terminal 2M and driver-side side air bag module terminal A, SAS control module terminal 2O and driver-side side air bag module terminal B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
5	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect the all SAS control module connectors. • Connect the driver-side side air bag module connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1992, B1993, B1994, and/or B1995 indicated?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	The system is normal at present. (Clear the malfunction from the memory.)

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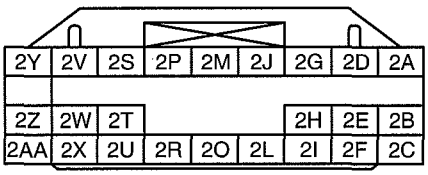
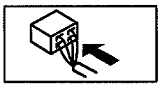
DTC B1996, B1997, B1998, B1999

dcf080200000w17

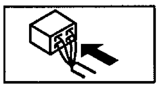
DTC	B1996	**Passenger-side side air bag module circuit short to power supply**
	B1997	**Passenger-side side air bag module circuit short to ground**
	B1998	**Passenger-side side air bag module circuit resistance high**
	B1999	**Passenger-side side air bag module circuit resistance low**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Resistance other than **1.4—3.2 ohms** detected in passenger-side side air bag module circuit • Malfunction in wiring harness between passenger-side side air bag module and SAS control module
POSSIBLE CAUSE		• Open or short circuit in wiring harness between passenger-side side air bag module and SAS control module • Passenger-side side air bag module malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR





PASSENGER-SIDE SIDE AIR BAG MODULE WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT PASSENGER-SIDE SIDE AIR BAG MODULE** • Using the current diagnostic tool, verify the following PID/DATA monitor. (See 08-02-6 PID/DATA MONITOR TABLE.) — PS_AB • Is the resistance of the passenger-side side air bag module normal? — Resistance: **1.4—3.2 ohms**	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
2	**INSPECT PASSENGER-SIDE SIDE AIR BAG MODULE CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the passenger-side side air bag module. • Disconnect the passenger-side side air bag module connector. • Is there any malfunction of the passenger-side side air bag module connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

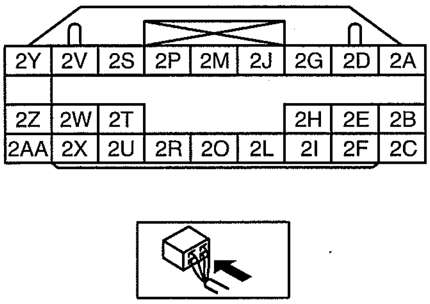
STEP	INSPECTION		ACTION
3	**VERIFY WHETHER MALFUNCTION IS IN PASSENGER-SIDE SIDE AIR BAG MODULE OR RELATED WIRING HARNESS** • Connect the leads of the **SST** (Fuel and thermometer checker) or apply **2-ohm** resistance to passenger-side side air bag module connector terminals A and B. • Set the resistance of the **SST** (Fuel and thermometer checker) to the **2-ohm** position. • Connect the negative battery cable. • Turn the engine switch to the ON position. • Are DTCs B1996, B1997, B1998 and/or B1999 indicated?	Yes	Go to the next step.
		No	Replace the passenger-side side air bag module. (See 08-10-5 SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
4	**INSPECT WIRING HARNESS BETWEEN PASSENGER-SIDE SIDE AIR BAG MODULE AND SAS CONTROL MODULE** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more.** • Disconnect the passenger-side side air bag module connector. • Remove the console. • Disconnect the SAS control module connectors. • Inspect the wiring harness between SAS control module terminal 2L and passenger-side side air bag module terminal A, SAS control module terminal 2I and passenger-side side air bag module terminal B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Go to the next step.
		No	Replace the air bag wiring harness.
5	**INSPECT AIR BAG SYSTEM WARNING LIGHT** • Connect the all SAS control module connectors. • Connect the passenger-side side air bag module connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Are DTCs B1996, B1997, B1998 and/or B1999 indicated?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	The system is normal at present. (Clear the malfunction from the memory.)

DTC B2444, U2017

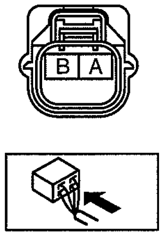
dcf080200000w18

DTC	B2444	**Driver-side side air bag sensor system internal circuit disabled**
	U2017	**Driver-side side air bag sensor system communication error**
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Malfunction in wiring harness between driver-side side air bag sensor and SAS control module • Malfunction in driver-side side air bag sensor circuit
POSSIBLE CAUSE		• Open or short circuit in wiring harness between driver-side side air bag sensor and SAS control module • Driver-side side air bag sensor malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



DRIVER-SIDE SIDE AIR BAG SENSOR WIRING HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT DRIVER-SIDE SIDE AIR BAG SENSOR CONNECTOR** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Disconnect the driver-side side air bag sensor connector. • Is there any malfunction of the driver-side side air bag sensor connector?	Yes	Replace the air bag wiring harness.
		No	Go to the next step.

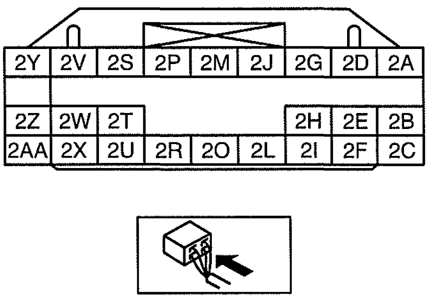
STEP	INSPECTION		ACTION
2	**INSPECT WIRING HARNESS BETWEEN DRIVER-SIDE SIDE AIR BAG SENSOR AND SAS CONTROL MODULE** • Turn the engine switch to the LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the console. • Disconnect the SAS control module connector. • Disconnect the driver-side side air bag sensor connector. • Inspect the wiring harnesses between SAS control module terminal 2AA and driver-side side air bag sensor terminal A, SAS control module terminal 2Z and driver-side side air bag sensor terminal B for the following: — Short to ground — Short to power supply — Open circuit • Is the wiring harness normal?	Yes	Replace the driver-side side air bag sensor, then go to the next step. (See 08-10-6 SIDE AIR BAG SENSOR REMOVAL/ INSTALLATION.)
		No	Replace the air bag wiring harness.
3	**INSPECT SAS CONTROL MODULE** • Connect the SAS control module connector. • Connect the driver-side side air bag sensor connector. • Connect the negative battery cable. • Turn the engine switch to the ON position. • Are DTCs B2444, U2017 indicated?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	DTC troubleshooting completed.

DTC B2445, U2018

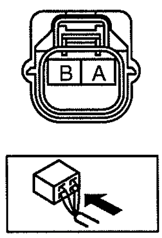
dcf080200000w19

DTC	B2445	Passenger-side side air bag sensor system internal circuit disabled
	U2018	Passenger-side side air bag sensor system communication error
DETECTION CONDITION		**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection according to only the detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • Malfunction in wiring harness between passenger-side side air bag sensor and SAS control module • Malfunction in passenger-side side air bag sensor circuit
POSSIBLE CAUSE		• Open or short circuit in wiring harness between passenger-side side air bag sensor and SAS control module • Passenger-side side air bag sensor malfunction • SAS control module malfunction

SAS CONTROL MODULE WIRING HARNESS-SIDE CONNECTOR



PASSENGER-SIDE SIDE AIR BAG SENSOR WIRING HARNESS-SIDE CONNECTOR



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Diagnostic procedure

<table>
<tr><th>STEP</th><th colspan="2">INSPECTION</th><th>ACTION</th></tr>
<tr><td rowspan="2">1</td><td rowspan="2">INSPECT PASSENGER-SIDE SIDE AIR BAG SENSOR CONNECTOR

Warning
• Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.
(See 08-10-2 SERVICE WARNINGS.)
(See 08-10-3 SERVICE CAUTIONS.)

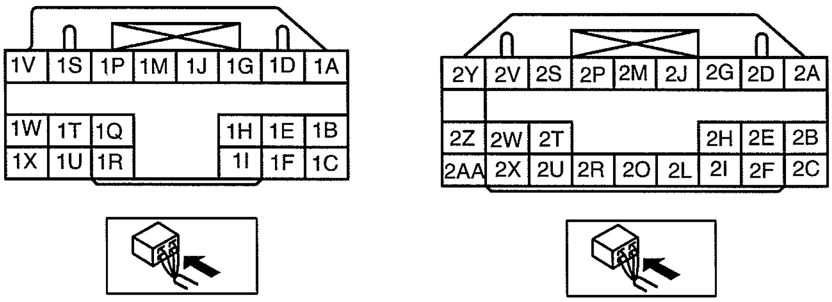
• Turn the engine switch to the LOCK position.
• Disconnect the negative battery cable and wait for 1 min or more.
• Disconnect the passenger-side side air bag sensor connector.
• Is there any malfunction of the passenger-side side air bag sensor connector?</td><td>Yes</td><td>Replace the air bag wiring harness.</td></tr>
<tr><td>No</td><td>Go to the next step.</td></tr>
<tr><td rowspan="2">2</td><td rowspan="2">INSPECT WIRING HARNESS BETWEEN PASSENGER-SIDE SIDE AIR BAG SENSOR AND SAS CONTROL MODULE
• Turn the engine switch to the LOCK position.
• Disconnect the negative battery cable and wait for 1 min or more.
• Remove the console.
• Disconnect the SAS control module connector.
• Disconnect the passenger-side side air bag sensor connector.
• Inspect the wiring harnesses between SAS control module terminal 2C and passenger-side side air bag sensor terminal A, SAS control module terminal 2B and passenger-side side air bag sensor terminal B for the following:
— Short to ground
— Short to power supply
— Open circuit
• Is the wiring harness normal?</td><td>Yes</td><td>Replace the passenger-side side air bag sensor, then go to the next step.
(See 08-10-6 SIDE AIR BAG SENSOR REMOVAL/ INSTALLATION.)</td></tr>
<tr><td>No</td><td>Replace the air bag wiring harness.</td></tr>
<tr><td rowspan="2">3</td><td rowspan="2">INSPECT SAS CONTROL MODULE
• Connect the SAS control module connector.
• Connect the passenger-side side air bag sensor connector.
• Connect the negative battery cable.
• Turn the engine switch to the ON position.
• Are DTCs B2445, U2018 indicated?</td><td>Yes</td><td>Replace the SAS control module.
(See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)</td></tr>
<tr><td>No</td><td>DTC troubleshooting completed.</td></tr>
</table>

DTC B2867

dcf080200000w20

DTC B2867	SAS control module connector poor connection
DETECTION CONDITION	**Warning** • **Detection conditions are for understanding the DTC outline before performing an inspection. Performing an inspection with only detection conditions may cause injury due to an operating error, or damage the system. When performing an inspection, always follow the inspection procedure.** • There is no continuity between poor connection detector bar terminals of SAS control module.
POSSIBLE CAUSE	• Poor connection of any SAS control module connectors • Malfunction of any SAS control module connectors • SAS control module malfunction

SAS CONTROL MODULE WIRING
HARNESS-SIDE CONNECTOR



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**VERIFY THAT ALL SAS CONTROL MODULE CONNECTORS ARE CONNECTED WITH SAS CONTROL MODULE** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the console. • Are all SAS control module connectors securely connected?	Yes	Go to the next step.
		No	Reconnect connector properly.

STEP	INSPECTION		ACTION
2	**INSPECT ALL SAS CONTROL MODULE CONNECTORS** • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. • Disconnect the passenger-side air bag module connector. • Disconnect the driver and passenger-side side air bag module connectors. (Vehicles with side air bag) • Disconnect the driver and passenger-side seat connectors. (Vehicles with side air bag) • Disconnect the driver and passenger-side pre-tensioner seat belt connectors. (Vehicle with pre-tensioner seat belt) • Remove the console. • Disconnect the all SAS control module connectors. • Are poor connection detector bars of all SAS control module connectors normal?	Yes	Replace the SAS control module. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Replace wiring harnesses.

08-03 SYMPTOM TROUBLESHOOTING

AIR BAG SYSTEM

dcf080300000w01

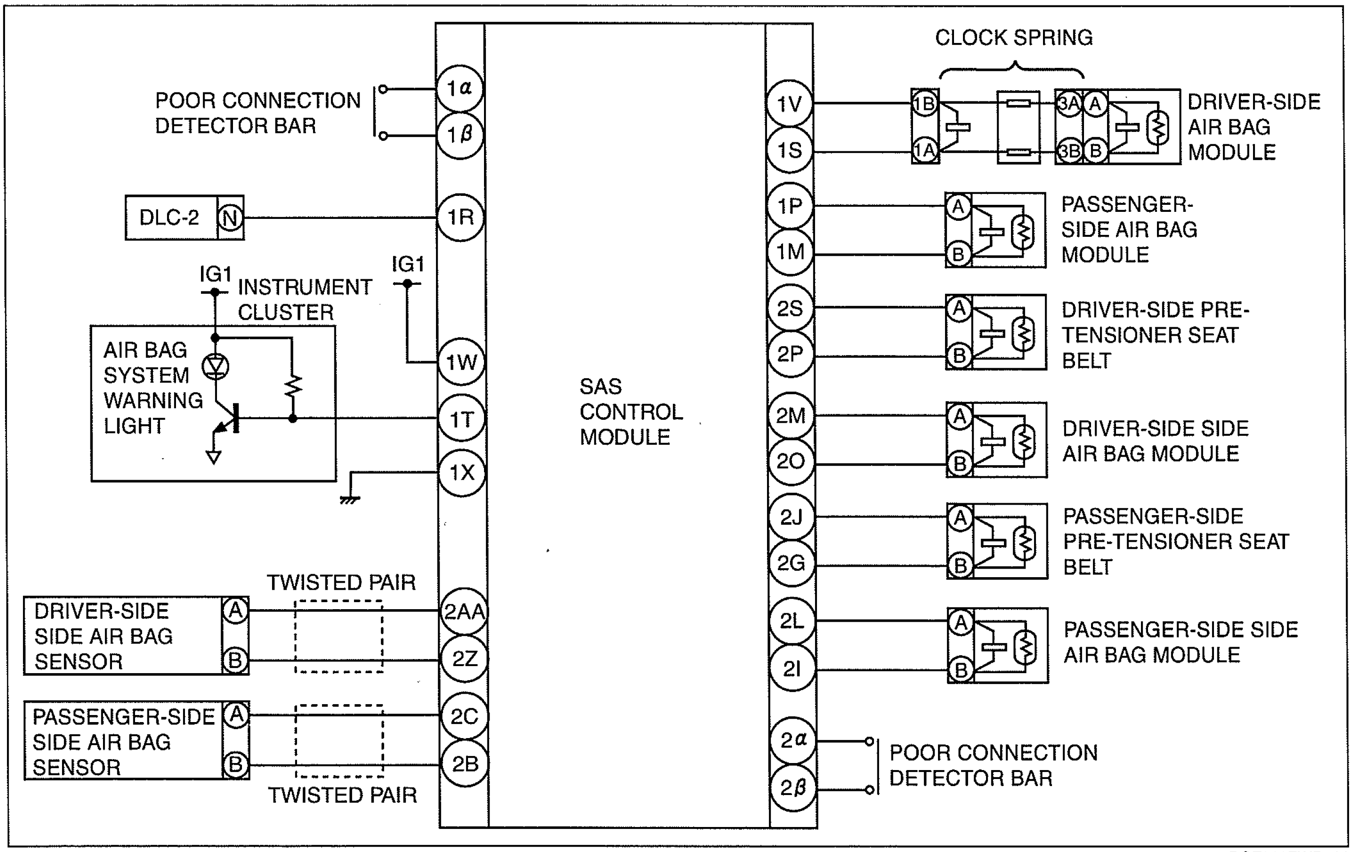
Troubleshooting Index

- Use the chart below verify the symptoms of the trouble in order to diagnose the appropriate area.

No.	Troubleshooting item	Description	Page
1	Air bag system warning light does not illuminate.	Malfunction in air bag system warning light circuit (short to ground).	(See 08-03-1 NO.1 AIR BAG SYSTEM WARNING LIGHT DOES NOT ILLUMINATE)
2	Air bag system warning light is illuminated constantly.	Malfunction in air bag system warning light circuit (open circuit or short to power supply).	(See 08-03-3 NO.2 AIR BAG SYSTEM WARNING LIGHT ILLUMINATES CONSTANTLY)

AIR BAG SYSTEM WIRING DIAGRAM (SYMPTOM TROUBLESHOOTING)

dcf080300000w02



DCF810ZWB001

NO.1 AIR BAG SYSTEM WARNING LIGHT DOES NOT ILLUMINATE

dcf080300000w03

1	Air bag system warning light does not illuminate.
DETECTION CONDITION	Malfunction in air bag system warning light circuit (short to ground)
POSSIBLE CAUSE	• SAS control module malfunction • Instrument cluster (circuit board) malfunction • Short to ground circuit in wiring harness between instrument cluster and SAS control module

Diagnostic Procedure

- When performing an asterisked (*) troubleshooting inspection, slightly shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunction. If there is a problem, verify that connectors, terminals and wiring harness are connected correctly and undamaged.

STEP	INSPECTION		ACTION
1	**INSPECT OTHER WARNING AND INDICATOR LIGHTS CIRCUIT IN INSTRUMENT CLUSTER** • Turn the engine switch to the ON position. • Do other warning and indicator lights illuminate?	Yes	Turn the engine switch to the LOCK position, then go to the next step.
		No	Inspect instrument cluster power supply system and ground system the, then go to Step 4.
2	**INSPECT SAS CONTROL MODULE** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the air bag system service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. • Disconnect the passenger-side air bag module connector. • Disconnect the driver and passenger-side seat connectors. • Disconnect the driver- and passenger-side pre-tensioner seat belt connectors. • Remove the console. • Disconnect all SAS control module connectors. • Connect the negative battery cable. • Turn the engine switch to ON position. • Does the air bag system warning light illuminate?	Yes	Replace the SAS control module, then go to Step 4. (See 08-10-6 SAS CONTROL MODULE REMOVAL/ INSTALLATION.)
		No	Go to the next step.
*3	**INSPECT WIRING HARNESS BETWEEN SAS CONTROL MODULE AND INSTRUMENT CLUSTER FOR SHORT TO GROUND** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable. • Disconnect the instrument cluster connector. • Is there continuity between terminal 1H of the instrument cluster connector and ground?	Yes	Replace the wiring harness, then go to Step 4.
		No	Replace the instrument cluster, then go to the next step. (See 09-22-2 INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
4	**CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Connect all SAS control module connectors. • Connect the driver and passenger-side pre-tensioner seat belt connectors. • Connect the driver and passenger-side seat connectors. • Connect the passenger-side air bag module connector. • Connect the clock spring connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Does the air bag system warning light operate properly?	Yes	Complete troubleshooting, then explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

NO.2 AIR BAG SYSTEM WARNING LIGHT ILLUMINATES CONSTANTLY

dcf080300000w04

2	**Air bag system warning light is illuminated constantly.**
DETECTION CONDITION	Malfunction in air bag system warning light circuit (open circuit or short to power supply).
POSSIBLE CAUSE	• Weak battery • SAS control module malfunction • Instrument cluster (circuit board) malfunction • No connection in SAS control module connector • Poor contact in instrument cluster connector (16-pin) • Open or short to power supply circuit in wiring harness between instrument cluster and SAS control module • Poor contact at terminals 1X and/or 1W of SAS control module connector • Poor contact in wiring harness between terminal 1X of SAS control module connector and ground • Poor contact in wiring harness between battery and terminal 1W of SAS control module

Diagnostic Procedure

- When performing an asterisked (*) troubleshooting inspection, slightly shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunction. If there is a problem, verify that connectors, terminals and wiring harness are connected correctly and undamaged.

STEP	INSPECTION		ACTION
1	**INSPECT BATTERY** • Measure the voltage of battery. • Is the voltage **9 V or more**?	Yes	Go to the next step.
		No	Battery is weak. Inspect charge/discharge system, then go to Step 9. (See 01-17-2 BATTERY INSPECTION.)
2	**VERIFY THAT SAS CONTROL MODULE CONNECTOR IS CONNECTED** **Warning** • **Handling the air bag system components improperly can accidentally deploy the air bag modules and pre-tensioner seat belts, which may seriously injure you. Read the air bag system service warnings and cautions before handling the air bag system components.** **(See 08-10-2 SERVICE WARNINGS.)** **(See 08-10-3 SERVICE CAUTIONS.)** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Remove the console. • Are all SAS control module connectors securely connected?	Yes	Go to the next step.
		No	Reconnect the connector properly, then go to Step 9.
* 3	**INSPECT WIRING HARNESS BETWEEN SAS CONTROL MODULE AND INSTRUMENT CLUSTER FOR CONTINUITY** • Remove the column cover. • Disconnect the clock spring connector. • Remove the glove compartment. • Disconnect the passenger-side air bag module connector. • Disconnect the driver and passenger-side seat connectors. • Disconnect the driver- and passenger-side pre-tensioner seat belt connectors. • Remove the console. • Disconnect all SAS control module connectors. • Disconnect the instrument cluster connector. • Is there continuity between SAS control module connector (20-pin) terminal 1T and instrument cluster connector terminal 1H?	Yes	Go to the next step.
		No	Replace the wiring harness, then go to Step 9.

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<table>
<tr><th>STEP</th><th colspan="2">INSPECTION</th><th>ACTION</th></tr>
<tr><td rowspan="2">* 4</td><td rowspan="2">INSPECT WIRING HARNESS BETWEEN SAS CONTROL MODULE AND INSTRUMENT CLUSTER FOR SHORT TO POWER SUPPLY
• Connect the negative battery cable.
• Turn the engine switch to ON position.
• Measure the voltage at instrument cluster connector terminal 1H.
• Is the voltage 9 V or more?</td><td>Yes</td><td>Replace the wiring harness, then go to Step 9.</td></tr>
<tr><td>No</td><td>Go to the next step.</td></tr>
<tr><td rowspan="2">5</td><td rowspan="2">CHECK TO SEE WHETHER MALFUNCTION IS IN AIR BAG SYSTEM WARNING LIGHT IN INSTRUMENT CLUSTER
• Connect instrument cluster connector terminal 1H to ground, then reconnect the connector
• Does the air bag system warning light illuminate with engine switch ON?</td><td>Yes</td><td>Replace the instrument cluster, then go to Step 9.
(See 09-22-2 INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)</td></tr>
<tr><td>No</td><td>Go to the next step.</td></tr>
<tr><td rowspan="2">6</td><td rowspan="2">INSPECT POWER SUPPLY CIRCUIT OF SAS CONTROL MODULE (TERMINAL (20-pin) 1W)
• Turn the engine switch to LOCK position.
• Disconnect the negative battery cable and wait for 1 min or more.
• Connect all SAS control module connectors.
• Connect the driver and passenger-side pre-tensioner seat belt connectors.
• Connect the driver and passenger-side seat connectors.
• Connect the passenger-side air bag module connector.
• Connect the clock spring connector.
• Inspect the voltage for PID/DATA monitor "IG_V_2" item using current diagnostic tool.
• Is the voltage of at least one terminal 9 V or more?</td><td>Yes</td><td>Go to the Step 8.</td></tr>
<tr><td>No</td><td>Go to the next step.</td></tr>
<tr><td rowspan="2">7</td><td rowspan="2">INSPECT WIRING HARNESS BETWEEN BATTERY AND FUSE BLOCK
• Connect the negative battery cable.
• Turn the engine switch to ON position.
• Measure the voltage at instrument cluster connector terminal 2A.
• Is the voltage 9 V or more?</td><td>Yes</td><td>Go to the next step.</td></tr>
<tr><td>No</td><td>Repair the wiring harnesses, then go to Step 9.</td></tr>
<tr><td rowspan="2">8</td><td rowspan="2">VERIFY THAT SAS CONTROL MODULE CONNECTOR TERMINAL 1X IS GROUND
• Turn the engine switch to LOCK position.
• Disconnect the negative battery cable and wait for 1 min or more.
• Remove the column cover.
• Disconnect the clock spring connector.
• Remove the glove compartment.
• Disconnect the passenger-side air bag module connector.
• Disconnect the driver and passenger-side side air bag module connectors.
• Disconnect the driver and passenger-side pre-tensioner seat belt connectors.
• Remove the console.
• Disconnect all SAS control module control model connectors.
• Inspect the wiring harness between SAS control module connector (20-pin) terminal 1X and ground for the following:
— Short to power supply
— Open circuit
• Is the wiring harness normal?</td><td>Yes</td><td>Replace the SAS control module, then go to the next step.
(See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)</td></tr>
<tr><td>No</td><td>Replace the wiring harnesses, then go to the next step.</td></tr>
</table>

STEP	INSPECTION		ACTION
9	**CONFIRM THAT MALFUNCTION SYMPTOMS DO NOT RECUR AFTER REPAIR** • Turn the engine switch to LOCK position. • Disconnect the negative battery cable and wait for **1 min or more**. • Connect all SAS control module connectors. • Connect the driver and passenger-side pre-tensioner seat belt connectors. • Connect the driver and passenger-side seat connectors. • Connect the passenger-side air bag module connector. • Connect the clock spring connector. • Connect the instrument cluster connector. • Connect the negative battery cable. • Turn the engine switch to ON position. • Does the air bag system warning light operate properly?	Yes	Complete troubleshooting, then explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.

08-10 AIR BAG SYSTEM

LOCATION INDEX

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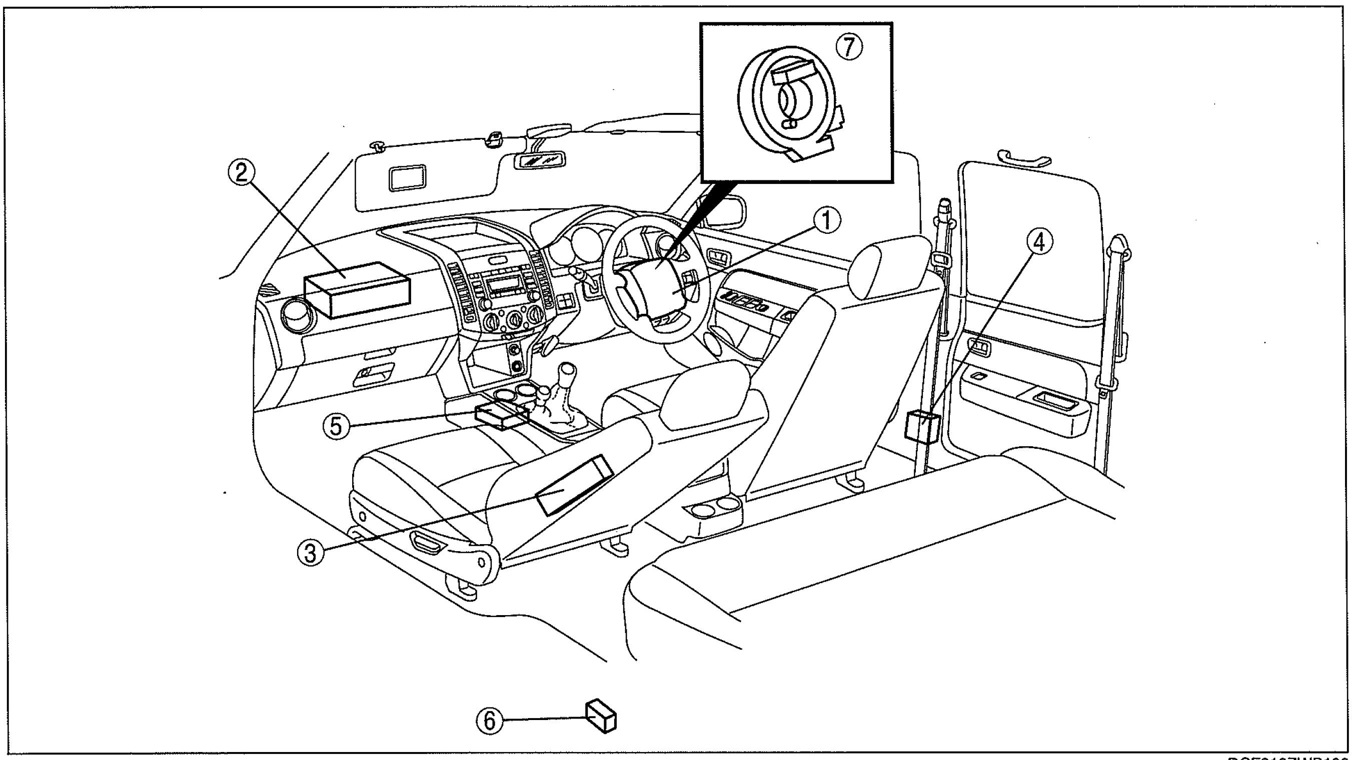


DCF810ZWB100

1	Driver-side air bag module (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION.) (See 08-10-9 AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES.)
2	Passenger-side air bag module (See 08-10-5 PASSENGER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION.) (See 08-10-9 AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES.)
3	Side air bag module (See 08-10-5 SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.) (See 08-10-9 AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES.)

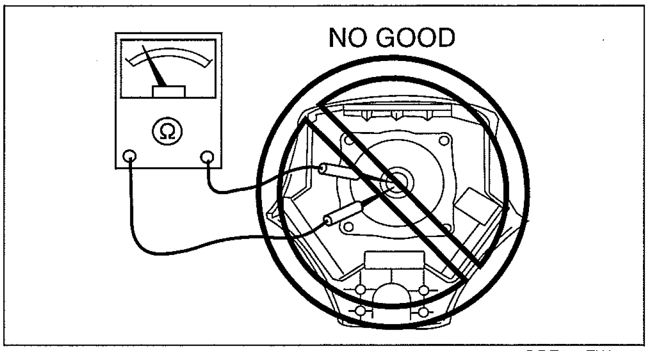
4	Pre-tensioner seat belt (See 08-10-9 AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES.)
5	SAS control module (See 08-10-6 SAS CONTROL MODULE REMOVAL/INSTALLATION.)
6	Side air bag sensor (See 08-10-6 SIDE AIR BAG SENSOR REMOVAL/ INSTALLATION.)
7	Clock spring (See 08-10-7 CLOCK SPRING REMOVAL/ INSTALLATION.) (See 08-10-8 CLOCK SPRING INSPECTION.) (See 08-10-7 CLOCK SPRING ADJUSTMENT.)

SERVICE WARNINGS

dcf081000000w02

Air Bag Module Inspection

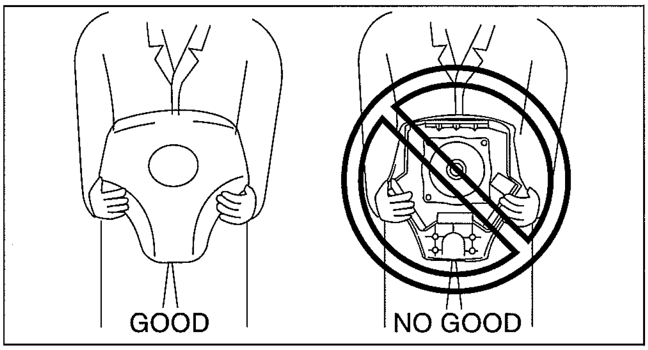
- Inspecting an air bag module using a tester can operate (deploy) the air bag module, which may cause serious injury. Do not use a tester to inspect an air bag module. Always use the on-board diagnostic function to diagnose the air bag module for malfunctions.



DPE810ZW1002

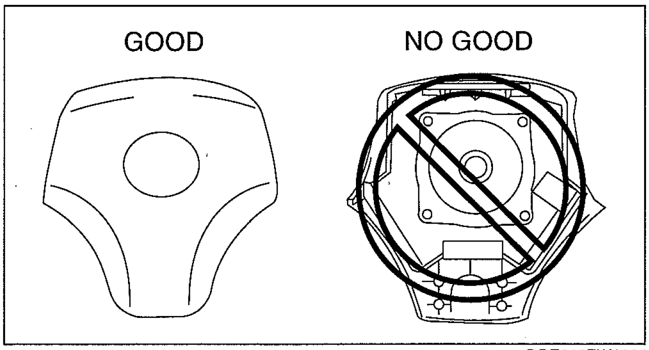
Air Bag Module Handling

- Before removing the air bag module or disconnecting the air bag module connector, always turn the ignition switch to the LOCK position, disconnect the negative battery cable, and then wait for 1 min or more to allow the backup power supply of the SAS control module to deplete its stored power.
- Handling a live (undeployed) air bag module that is pointed toward your body could result in serious injury if the air bag module were to accidentally operate (deploy). When carrying a live (undeployed) air bag module, point the deployment surface away from your body to lessen the chance of injury in case it operates (deploys).



DPE810ZW1003

- A live (undeployed) air bag module placed with its deployment surface to ground is dangerous. If the air bag module were to accidentally operate (deploy), it could cause serious injury. Always place a live (undeployed) air bag module with its deployment surface up.



DPE810ZW1004

Side Air Bag Module Handling

- When a side air bag module operates (deploys) due to a collision, the interior of the seat back (pad, frame, trim) may become damaged. If a side air bag does not operate (deploy) normally from a seat back that has been reused, a serious accident may result. After a side air bag has operated (deployed), always replace both the side air bag module and the seat back (pad, frame, trim) with new parts. After servicing, verify that the seat operates normally and that the wiring harness is not caught.

SAS control module Handling

- Removing the SAS control module or disconnecting the SAS control module connector with the ignition switch at the ON position can activate the sensor in the SAS control module and operate (deploy) the air bags and pre-tensioner seat belts, which may cause serious injury. Before removing the SAS control module or disconnecting the SAS control module connector, always turn the ignition switch to the LOCK position, disconnect the negative battery cable, and then wait for 1 min or more to allow the backup power supply of the SAS control module to deplete its stored power.
- Connecting the SAS control module connector with the SAS control module not securely fixed to the vehicle is dangerous. The sensor in the SAS control module could send an electrical signal to the air bag modules and pre-tensioner seat belts. This will operate (deploy) the air bags and pre-tensioner seat belts, which may result in serious injury. Therefore, before connecting the connector, securely fix the SAS control module to the vehicle.

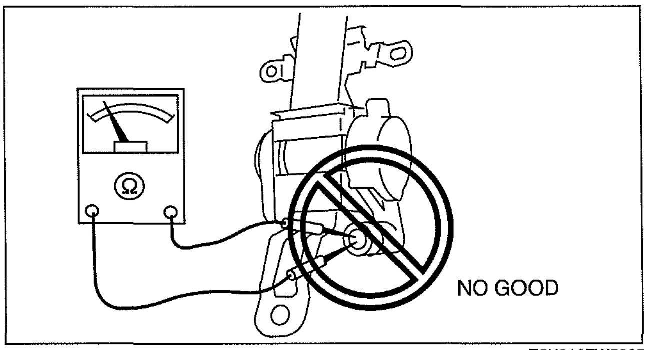
- Because a sensor is built into the SAS control module, once the air bags and pre-tensioner seat belts have operated (deployed) due to a collision or other causes, the SAS control module must be replaced with a new one even if the used one does not have any visible external damage or deformation. The used SAS control module may have been damaged internally which may cause improper operation. If the SAS control module is reused, the air bags and pre-tensioner seat belts may not operate (deploy) normally, which could result in a serious accident. Always replace the SAS control module with a new one. The SAS control module cannot be bench-checked or self-checked.

Side Air Bag Sensor Handling

- Removing the side air bag sensor or disconnecting the side air bag sensor connector with the ignition switch at the ON position can activate the side air bag sensor and operate (deploy) the side air bag, which may cause serious injury. Before removing the side air bag sensor or disconnecting the side air bag sensor connector, always turn the ignition switch to the LOCK position, disconnect the negative battery cable, and then wait for 1 min or more to allow the backup power supply of the SAS control module to deplete its stored power.
- If the side air bag sensor is subjected to shock or the sensor is disassembled, the side air bag may accidentally operate (deploy) and cause injury, or the system may fail to operate normally and cause a serious accident. Do not subject the side air bag sensor to shock or disassemble the sensor.
- Because a sensor is built into the side air bag sensor, once the air bag has operated (deployed) due to a collision or other causes, the side air bag sensor must be replaced with a new one even if the used one does not have any visible external damage or deformation. If the side air bag sensor is reused, the side air bag may not operate (deploy) normally, which could result in a serious accident. Always replace the side air bag sensor with a new one. The side air bag sensor cannot be bench-checked or self-checked.

Pre-tensioner Seat Belt Inspection

- Inspecting a pre-tensioner seat belt using a tester can operate (deploy) the pre-tensioner seat belt, which may cause serious injury. Do not use a tester to inspect a pre-tensioner seat belt. Always use the on-board diagnostic function to diagnose the pre-tensioner seat belt for malfunctions.



E5U810ZW5005

SERVICE CAUTIONS

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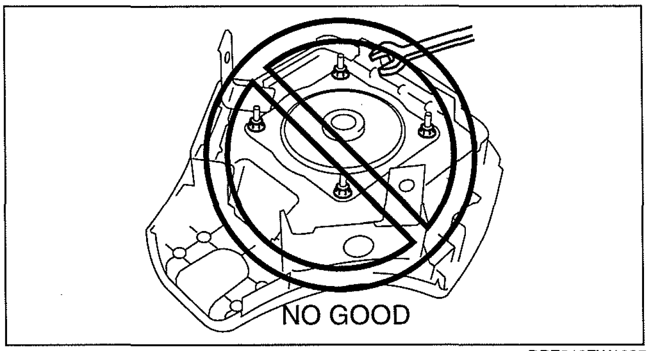
Air Bag System Component Disassembly

- Disassembling the air bag system components could cause it to not operate (deploy) normally. Never disassemble any air bag system components.



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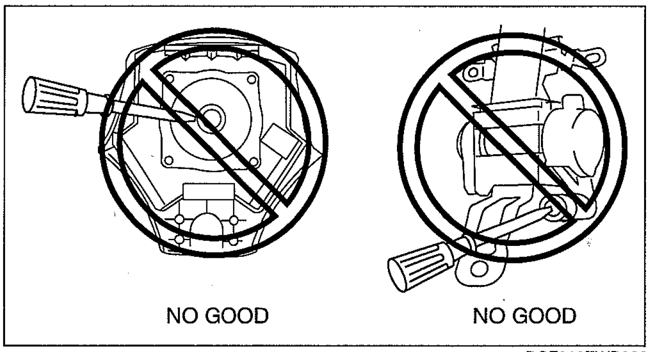
Air Bag Module, Pre-tensioner Seat Belt Handling

- **Oil, grease, or water on the air bag modules may cause the air bags and pre-tensioner seat belts to fail to operate (deploy) in an accident. Never allow oil, grease, or water to get on the air bag modules or pre-tensioner seat belts.**
- **Inserting a screwdriver or similar object into the connector of an air bag module or a pre-tensioner seat belt may damage the connector and cause the air bag module or the pre-tensioner seat belt to operate (deploy) improperly, which may cause serious injury. Never insert any foreign objects into the air bag module or seat belt connectors.**



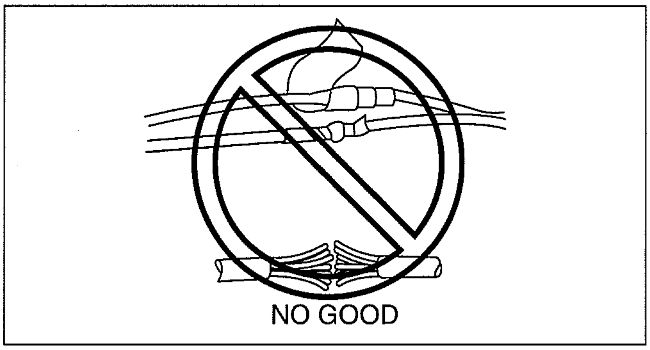
DCF810ZWB002

Air Bag Module, Pre-tensioner Seat Belt Reuse

- **Even if an air bag module or a pre-tensioner seat belt does not operate (deploy) in a collision and does not have any external signs of damage, it may have been damaged internally, which may cause improper operation. Before reusing a live (undeployed) air bag module and the pre-tensioner seat belts, always use the on-board diagnostic to diagnose the air bag module and the pre-tensioner seat belts to verify that they have no malfunction.**

Air Bag Wiring Harness Repair

- **Incorrectly repairing an air bag wiring harness can accidentally operate (deploy) the air bag module and pre-tensioner seat belts. If a problem is found in the air bag wiring harness, always replace the wiring harness with a new one.**



CHU0810W606

DRIVER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION

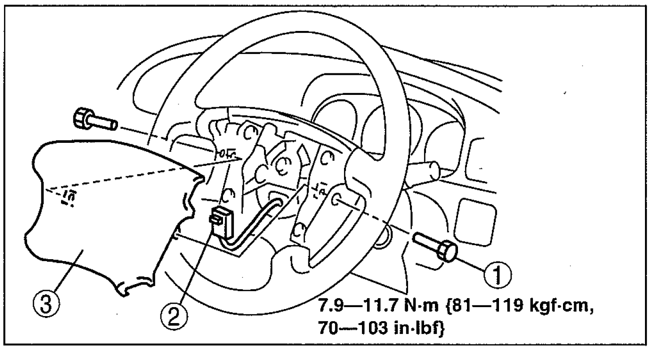
dcf081057010w01

Warning

- **Handling the air bag module improperly can accidentally deploy the air bag module, which may seriously injure you. Read the service warnings and cautions before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**

1. Turn the engine switch to the LOCK position.
2. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
3. Remove in the order indicated in the table.

1	Bolt
2	Connector (See 08-10-5 Connector Removal Note.)
3	Driver-side air bag module

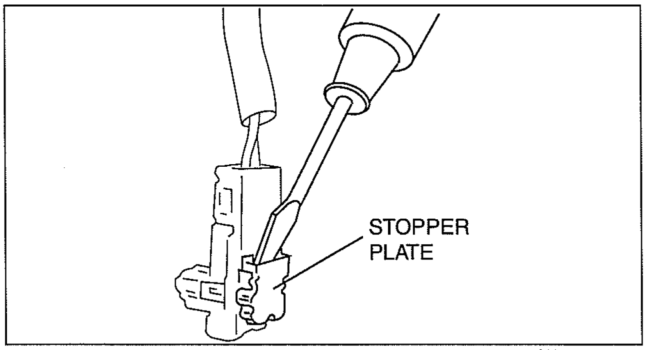


DCF810ZWB012

4. Install in the reverse order of removal.
5. Turn the engine switch to the ON position.
6. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

Connector Removal Note

1. Using a flathead screwdriver, pry out the connector stopper plate.
2. Disconnect the connector.



CHU0810W003

PASSENGER-SIDE AIR BAG MODULE REMOVAL/INSTALLATION

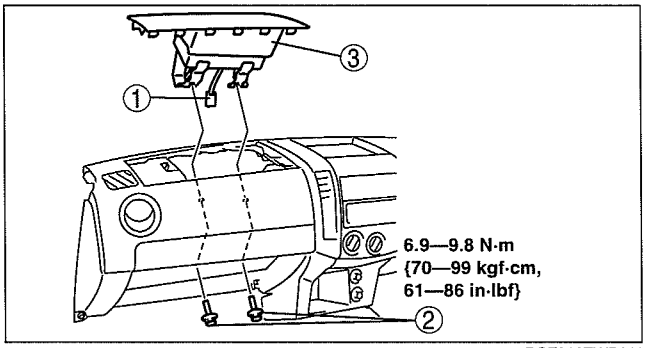
dcf081057050w01

Warning

- **Handling the air bag module improperly can accidentally deploy the air bag module, which may seriously injure you. Read the service warnings and cautions before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**

1. Turn the engine switch to the LOCK position.
2. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
3. Remove the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
4. Remove in the order indicated in the table.

1	Connector
2	Bolt
3	Passenger-side air bag module



DCF810ZWB013

5. Install in the reserve order of removal.
6. Turn the engine switch to the ON position.
7. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

SIDE AIR BAG MODULE REMOVAL/INSTALLATION

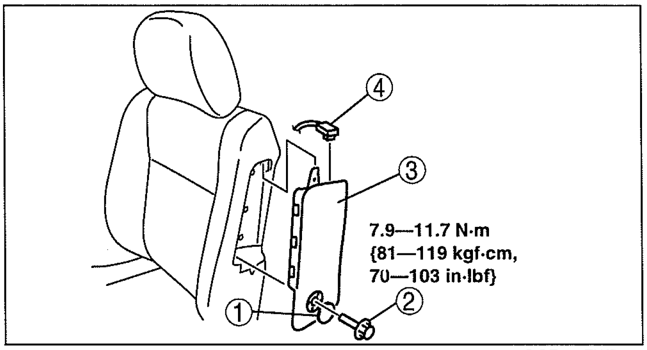
dcf081000147w01

Warning

- **Handling the air bag module improperly can accidentally operate (deploy) the air bag module, which may seriously injure you. Read the service warnings and cautions before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**
- **If the side air bag module is installed with debris in the seat back, the foreign material may be scattered when the side air bag module operates (deploys), causing injury. Verify that there is no foreign material in the seat back before installing the side air bag module.**

1. Turn the engine switch to the LOCK position.
2. Disconnect the negative battery cable and wait for **1 min or more.** (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
3. Remove in the order indicated in the table.

1	Cup
2	Bolt
3	Side air bag module
4	Connector



DCF810ZWB014

4. Install in the reverse order of removal.
5. Turn the engine switch to the ON position.
6. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

08

SAS CONTROL MODULE REMOVAL/INSTALLATION

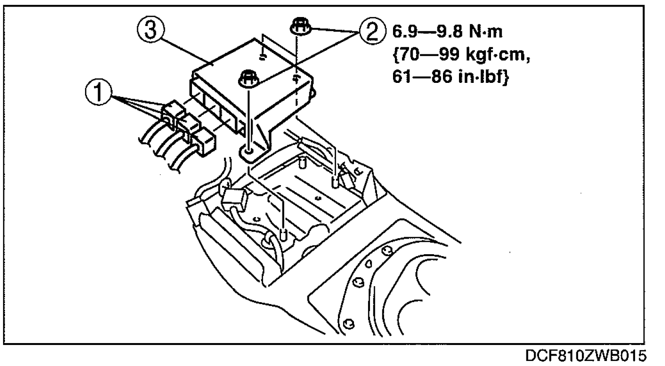
dcf081057030w01

Warning

- **Handling the SAS control module improperly can accidentally deploy the air bag modules and pretensioner seat belt, which may seriously injure you. Read the service warnings and cautions before handling the air bag module. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**

1. Turn the engine switch to the LOCK position.
2. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
3. Remove the console. (See 09-17-12 CONSOLE REMOVAL/INSTALLATION.)
4. Remove in the order indicated in the table.

1	Connector
2	Bolt
3	SAS control module



DCF810ZWB015

5. Install in the reverse order of removal.
6. Turn the engine switch to the ON position.
7. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

SIDE AIR BAG SENSOR REMOVAL/INSTALLATION

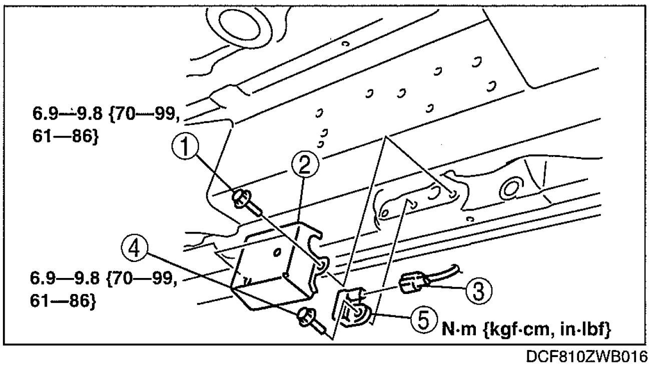
dcf081000146w01

Warning

- **Handling the side air bag sensor improperly can accidentally operate (deploy) the air bag module, which may seriously injure you. Read the service warnings and cautions before handling the side air bag sensor. (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**

1. Turn the engine switch to the LOCK position.
2. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
3. Remove in the order indicated in the table.

1	Bolt A
2	Cover
3	Connector
4	Bolt B
5	Side air bag sensor



DCF810ZWB016

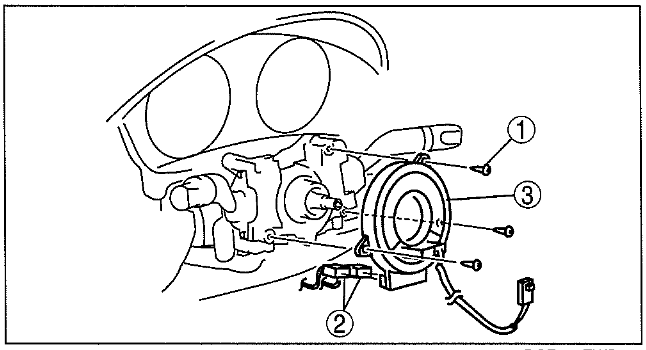
4. Install in the reverse order of removal.
5. Turn the engine switch to the ON position and hold for **5 s or more**.
6. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

CLOCK SPRING REMOVAL/INSTALLATION

dcf081066123w01

1. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
2. Remove the driver-side air bag module. (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
3. Remove the steering wheel. (See 06-14-6 STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
4. Remove the column cover. (See 09-17-8 COLUMN COVER REMOVAL/INSTALLATION.)
5. Remove in the order indicated in the table.

1	Screw
2	Connector
3	Clock spring (See 08-10-7 Clock Spring Installation Note.)



DCF810ZWB010

6. Install in the reverse order of removal.
7. Turn the ignition switch to the ON position.
8. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out.
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

Clock Spring Installation Note

Caution
- **If the clock spring is not adjusted, the spring wire in the clock spring could over-wind and break when the steering wheel is turned. Always adjust the clock spring after installing it.**

1. Adjust the clock spring after installing it. (See 08-10-7 CLOCK SPRING ADJUSTMENT.)

CLOCK SPRING ADJUSTMENT

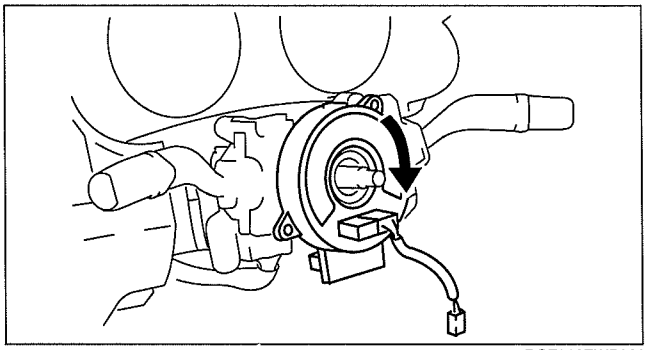
dcf081066123w02

Note
- The adjustment procedure is also specified on the caution label of the clock spring.

1. Set the front tires straight-ahead.

Caution
- **The clock spring will break if over-wound. Do not forcibly turn the clock spring.**

2. Turn the clock spring clockwise until it stops.


DCF810ZWB003

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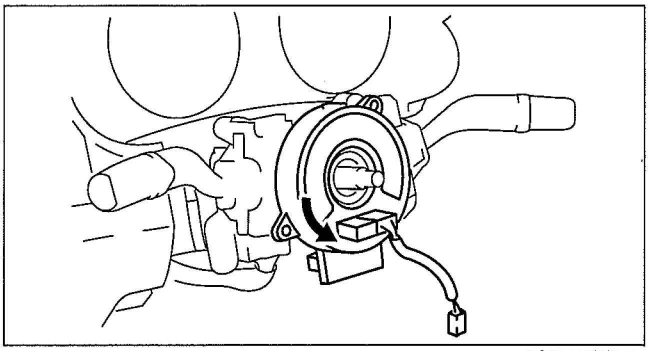
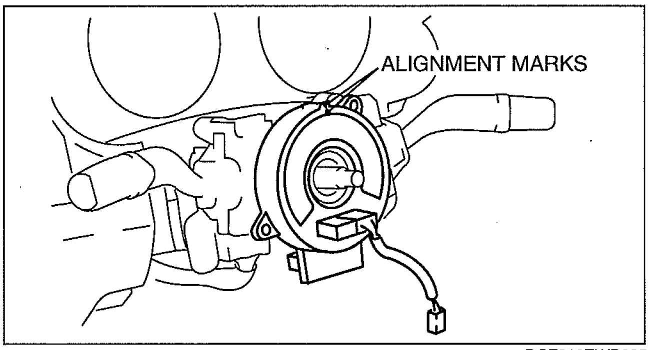
3. From the stopped position, turn the clock spring counterclockwise **2 3/4 turns.**



DCF810ZWB004

4. Align the mark on the clock spring with the mark on the outer housing.



DCF810ZWB005

CLOCK SPRING INSPECTION

dcf081066123w03

1. Verify that the continuity is as indicated in the table.
   - If not as indicated in the table, replace the clock spring.

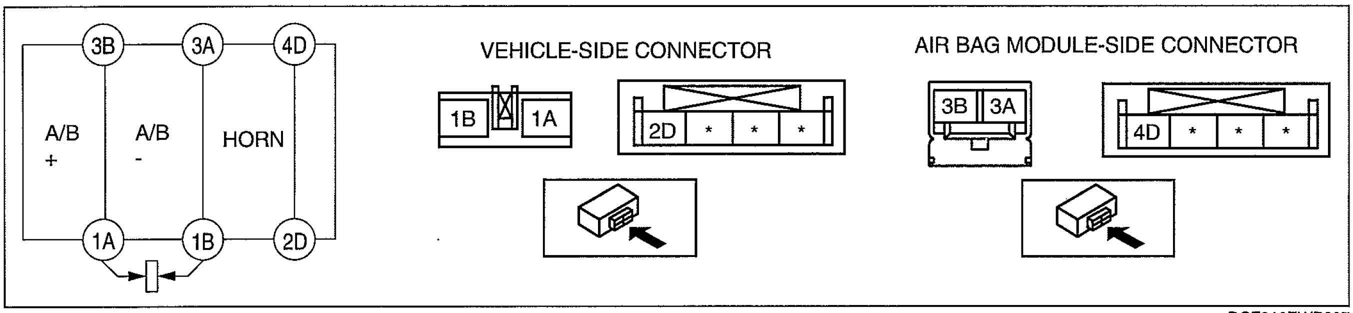
Note

- When the vehicle-side connector for the clock spring is disconnected, terminals 1A and 1B are shorted to prevent unexpected operation (deployment) of the air bag module.

O—O : Continuity

Test condition	Terminal 1A	1B	2D	3A	3B	4D
Under any condition	O				O	
		O		O		
			O			O

DCF810ZWB006



DCF810ZWB007

AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES

dcf081057000w01

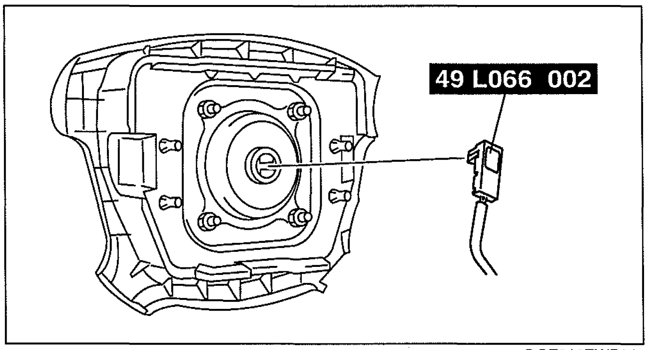
Warning
- **A live (undeployed) air bag module or pre-tensioner seat belt may accidentally operate (deploy) when it is disposed of and cause serious injury. Do not dispose of a live (undeployed) air bag module and pre-tensioner seat belt. If the SSTs (Deployment tool and Adapter harness) are not available, consult the nearest Mazda representative for assistance.**

Caution
- **Deploying the air bag modules and pre-tensioner seat belts inside the vehicle may cause damage to the vehicle interior. When the vehicle is not to be scrapped, always deploy the air bag modules and pre-tensioner seat belts outside the vehicle.**

- If the vehicle is to be scrapped, or when disposing of any air bag modules or pre-tensioner seat belts, operate (deploy) them inside the vehicle by following the deployment procedure below and using the **SST** (Deployment tool).
- When disposing of an operated (deployed) air bag module and pre-tensioner seat belt, refer to "AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DISPOSAL PROCEDURES".

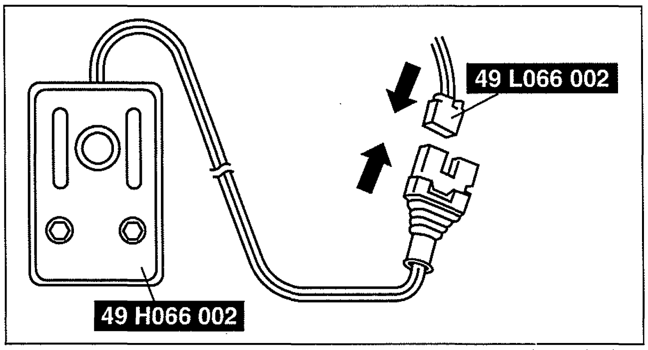
Deployment Procedure for Inside of Vehicle
1. Inspect the **SST** (Deployment tool). (See 08-10-19 INSPECTION OF SST (DEPLOYMENT TOOL).)
2. Move the vehicle to an open space, away from strong winds, and close all of the vehicle doors and windows.
3. Turn the engine switch to the LOCK position.
4. Disconnect the negative battery cable and wait for **1 min or more.** (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
5. Follow the procedure below for operating (deploying) the applicable air bag module or pre-tensioner seat belt.

Driver-side Air Bag Module
1. Remove the driver-side air bag module. (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
2. Connect the **SST** (Adapter harness) to the driver-side air bag module as shown in the figure.
3. Install the driver-side air bag module. (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)



DCF810ZWB017

4. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
5. Connect both **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
6. Verify that the red lamp on both **SST** (Deployment tool) is illuminated.
7. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.



CPJ810ZWB032

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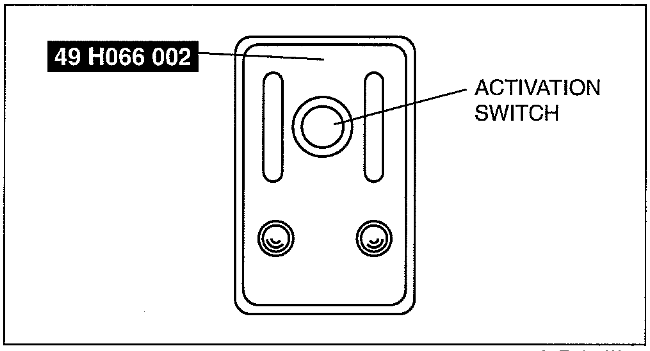
8. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the driver-side air bag module.

   Warning
   - **The air bag module is very hot immediately after it is operated (deployed). You can get burned. Do not touch the air bag module for at least 15 min after deployment.**

9. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).



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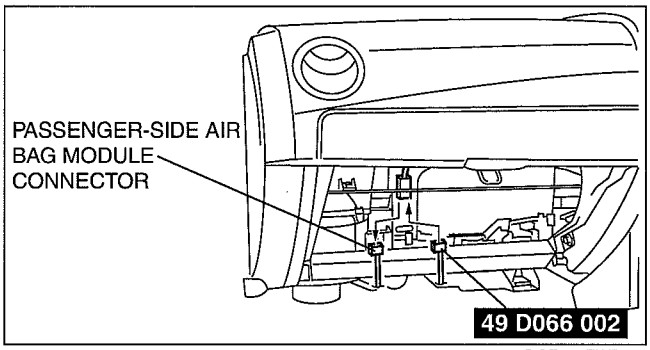
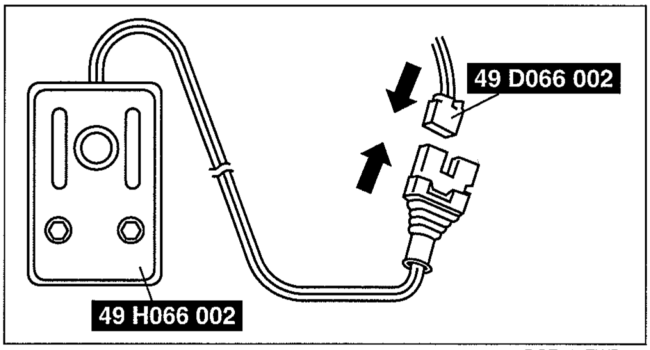
Passenger-side Air Bag Module

1. Remove the glove compartment. (See 09-17-9 GLOVE COMPARTMENT REMOVAL/INSTALLATION.)
2. Disconnect the passenger-side air bag module connector.
3. Connect the **SST** (Adapter harness) to the passenger-side air bag module as shown in the figure.



DCF810ZWB018

4. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
5. Connect the **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
6. Verify that the red lamp on both **SST** (Deployment tool) is illuminated.
7. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.

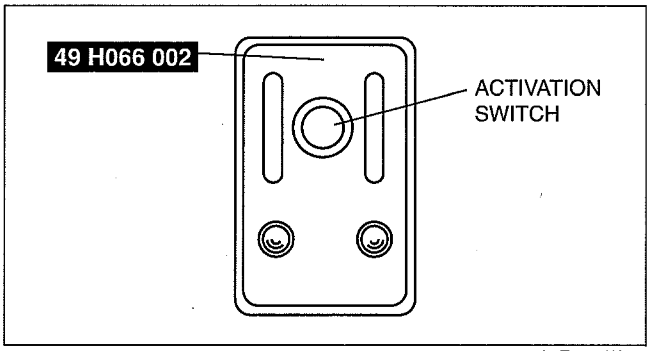


DCF810ZWB008

8. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the passenger-side air bag module.

   Warning
   - **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**

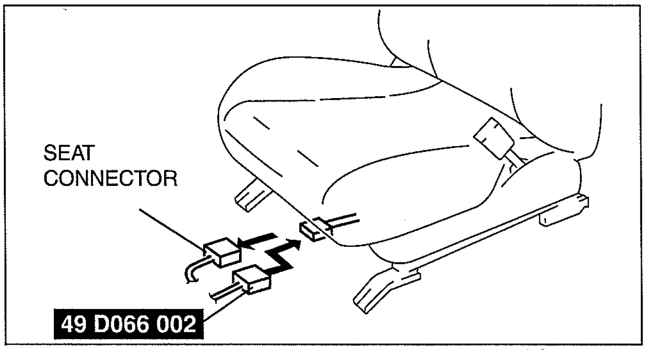
9. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).



A6E8130W028

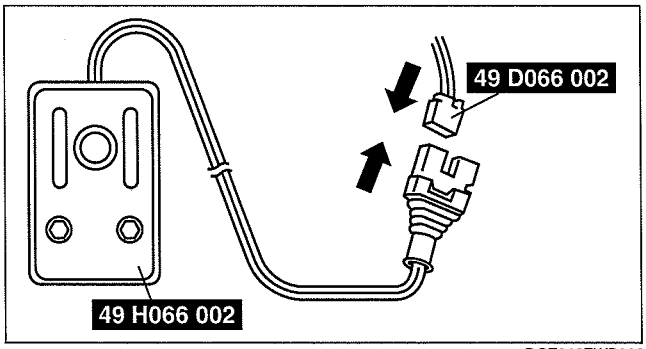
Side Air Bag Module

1. Disconnect the seat connector.
2. Connect the **SST** (Adapter harness) to the side air bag module as shown in the figure.



DCF810ZWB019

3. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
4. Connect the **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
5. Verify that the red lamp on the **SST** (Deployment tool) is illuminated.
6. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.

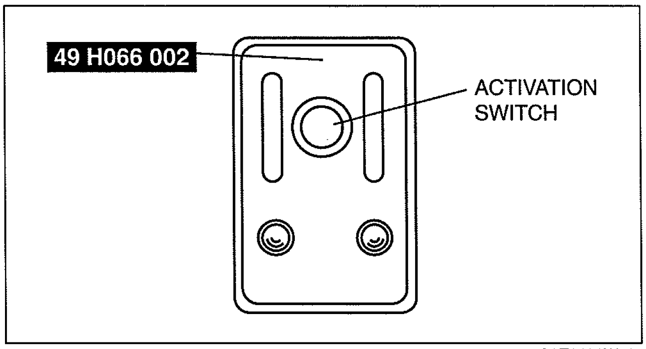


DCF810ZWB008

7. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the side air bag module.

   Warning
   - **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**

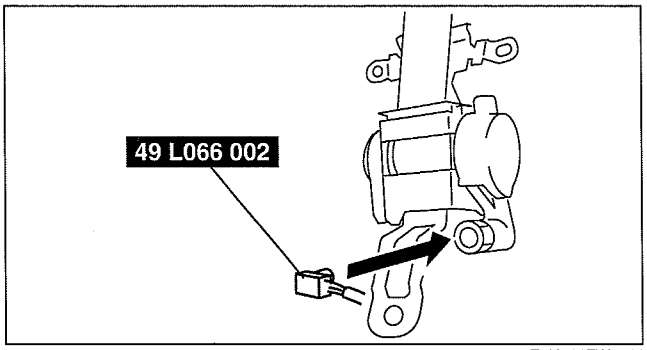
8. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).



A6E8130W028

Pre-tensioner Seat Belt

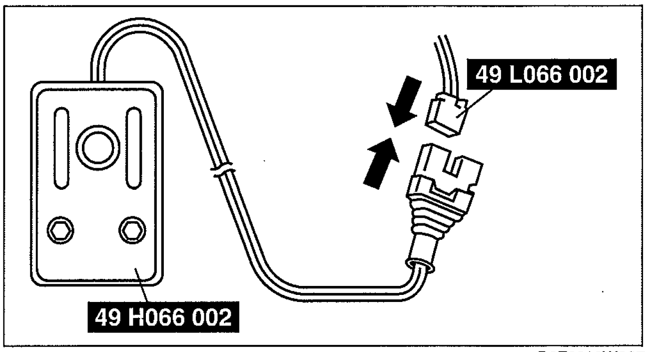
1. Remove the pre-tensioner seat belt and connect the **SST** (Adapter harness) as shown in the figure. (See 08-11-1 SEAT BELT REMOVAL/ INSTALLATION.)
2. Install the pre-tensioner seat belt.



E5U810ZW5026

08

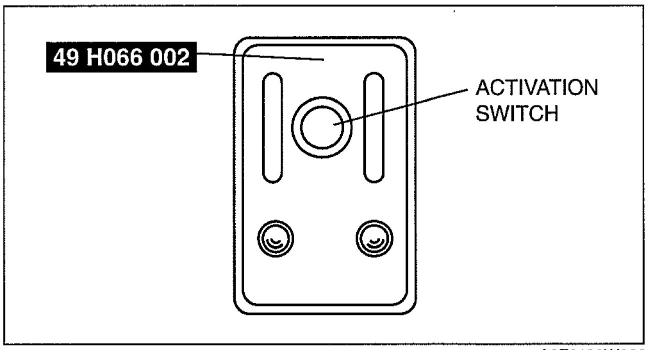
3. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
4. Connect the **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
5. Verify that the red lamp on the **SST** (Deployment tool) is illuminated.
6. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.



B3E0810W027

7. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the pre-tensioner seat belt.

Warning
- **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**

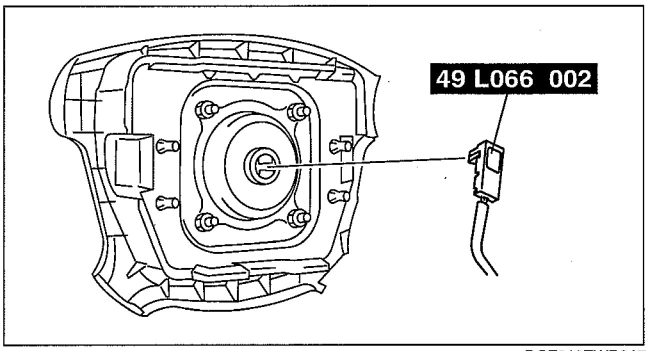


A6E8130W028

8. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).

Deployment Procedure for Outside of Vehicle
1. Inspect the **SST** (Deployment tool). (See 08-10-19 INSPECTION OF SST (DEPLOYMENT TOOL).)
2. Turn the engine switch to the LOCK position.
3. Disconnect the negative battery cable and wait for **1 min or more**. (See 01-17-2 BATTERY REMOVAL/ INSTALLATION.)
4. Follow the procedure below for operating (deploying) the applicable air bag module or pre-tensioner seat belt.

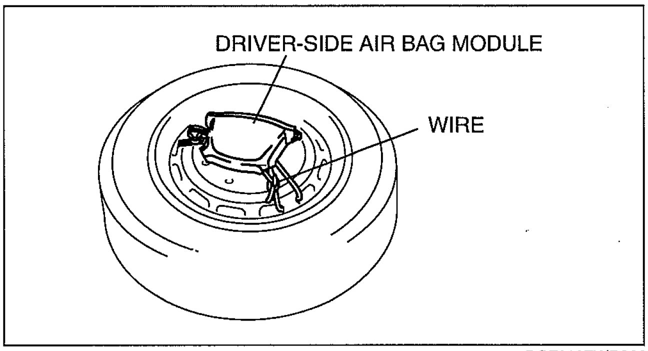
Driver-side Air Bag Module
1. Remove the driver-side air bag module. (See 08-10-4 DRIVER-SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
2. Connect the **SST** (Adapter harness) to the driver-side air bag module as shown in the figure.



DCF810ZWB017

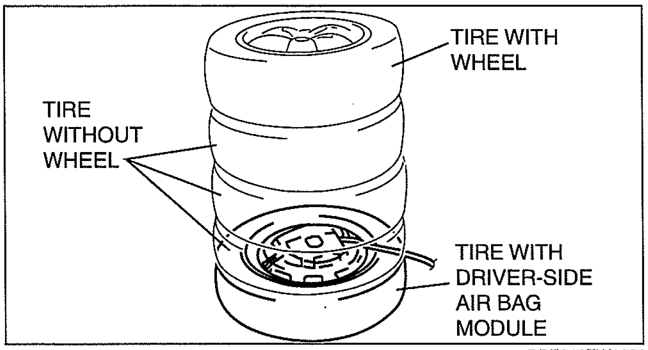
3. Place the driver-side air bag module on the center of the tire wheel with the padded surface facing up. To secure the air bag module to the tire wheel, wrap a wire (cross section **1.25 mm^2 {0.002 in^2} or more**) through the wheel and the bolt installation holes of the air bag module **at least 4 times**.

Warning
- **If the air bag module is not properly installed to the tire wheel, serious injury may occur when the module is operated (deployed). When installing the air bag module to the tire wheel, make sure the padded surface is facing up.**



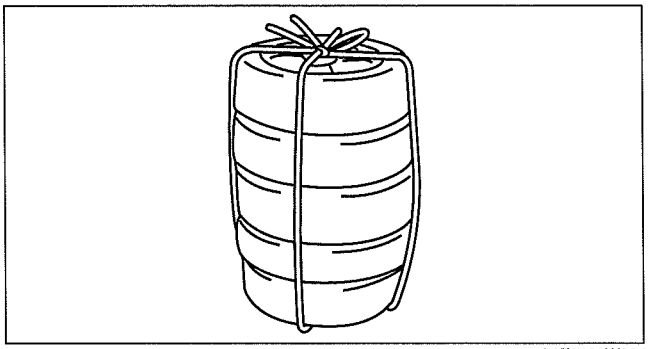
DCF810ZWB020

4. Stack three tires without wheels on top of the tire with the driver-side air bag module, and then stack another tire with a wheel on the very top.

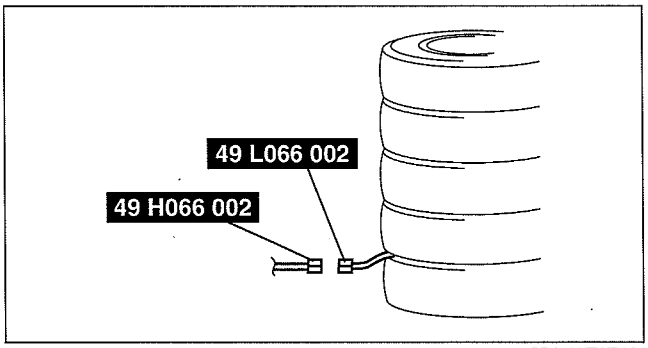


DPE810ZW1036

5. Secure the tires with wire.


A6E8130W034

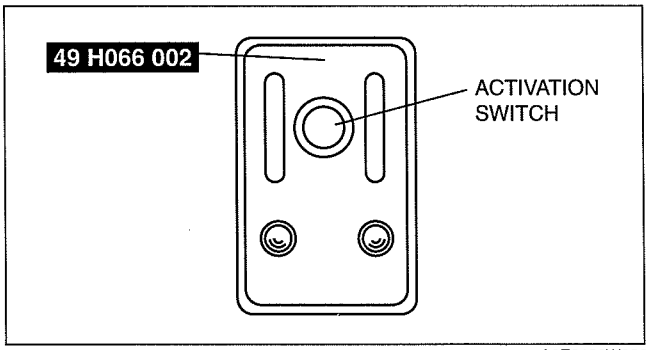
6. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
7. Connect both **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
8. Verify that the red lamp on both **SST** (Deployment tool) is illuminated.
9. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.



CPJ810ZWB034

10. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the driver-side air bag module.

   Warning
   - **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**



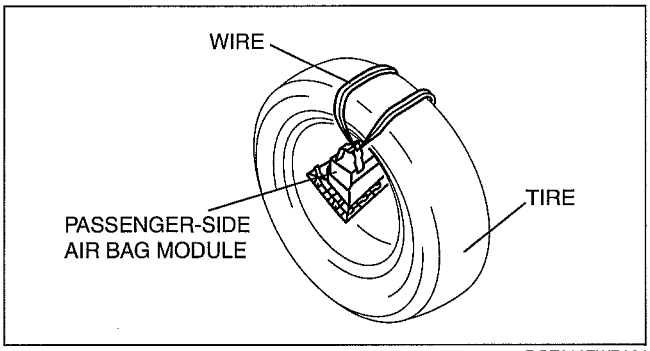
A6E8130W028

11. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).

Passenger-side Air Bag Module

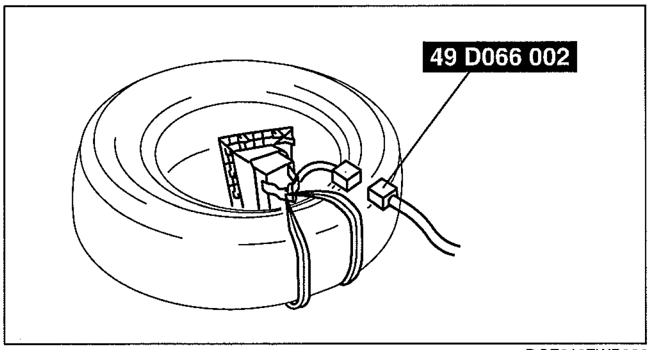
1. Remove the passenger-side air bag module. (See 08-10-5 PASSENGER-SIDE AIR BAG MODULE REMOVAL/ INSTALLATION.)
2. Place the padded surface of the passenger-side air bag module facing the center of the tire as shown in the figure. To secure the air bag module to the tire wheel, wrap a wire (cross section **1.25 mm^2 {0.002 in^2} or more**) through the tire and the bolt installation holes of the air bag module **at least 4 times.**

   Warning
   - **If the air bag module is not properly secured to the tire, the tires may fall over by the impact of operation (deployment) and cause serious injury. To prevent this, secure the air bag module properly with the padded surface facing the center of the tire.**



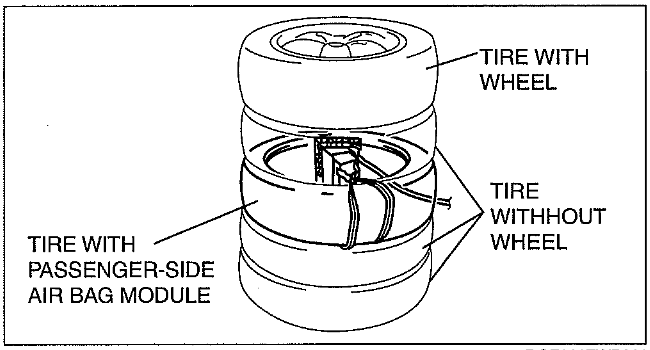
DCF810ZWB021

3. Connect the **SST** (Adapter harness) to the passenger-side air bag module as shown in the figure.



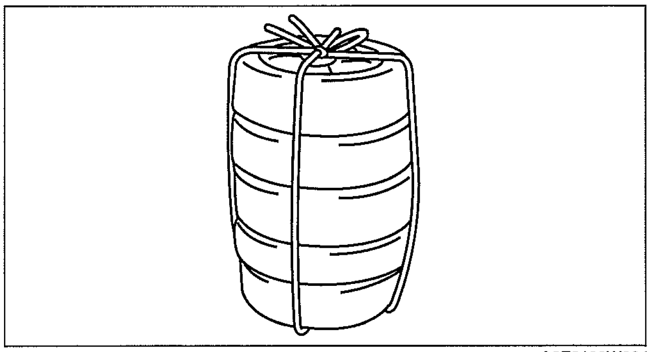
DCF810ZWB022

4. Stack the tire with the passenger-side air bag module on top of two tires without wheels. Stack a tire without a wheel on top of the tire with the passenger-side air bag module, and then stack another tire with a wheel on the very top.



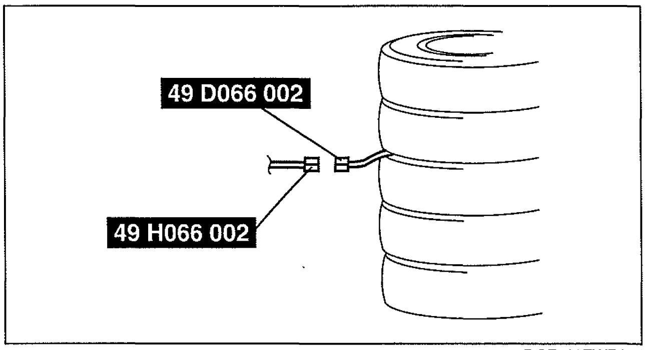
DCF810ZWB023

5. Secure the tires with wire.



A6E8130W034

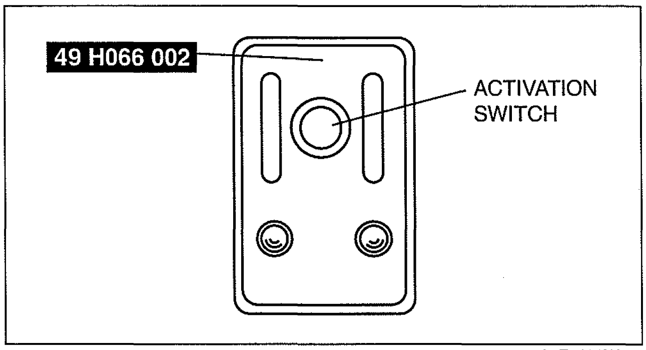
6. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
7. Connect both **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
8. Verify that the red lamp on both **SST** (Deployment tool) is illuminated.
9. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.



DCF810ZWB024

10. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the passenger-side air bag module.

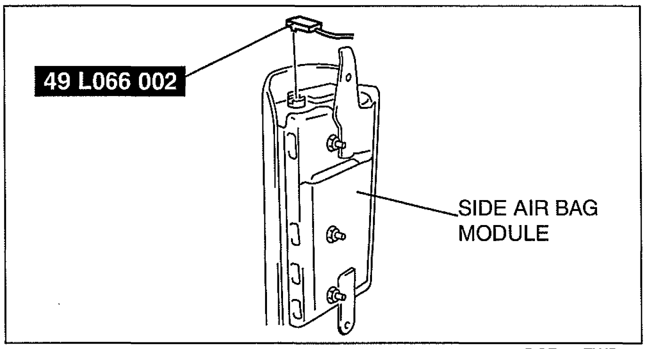
Warning
- **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**



A6E8130W028

11. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).

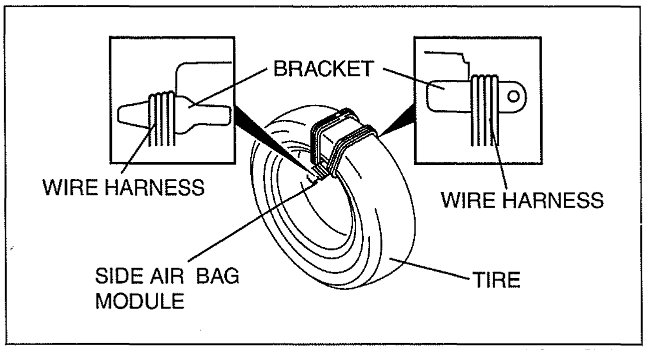
Side Air Bag Module
1. Remove the side air bag module. (See 08-10-5 SIDE AIR BAG MODULE REMOVAL/INSTALLATION.)
2. Connect the **SST** (Adapter harness) to the side air bag module as shown in the figure.



DCF810ZWB025

3. Place the padded surface of the side air bag module facing the center of the tire as shown in the figure. To secure the air bag module to the tire wheel, wrap a wire (cross section **1.25 mm^2 {0.002 in^2} or more**) through the tire and around the bracket **at least 4 times**.

Warning
- **If the air bag module is not properly secured to the tire, the tires may fall over by the impact of operation (deployment) and cause serious injury. To prevent this, secure the air bag module properly with the padded surface facing the center of the tire.**



DCF810ZWB026

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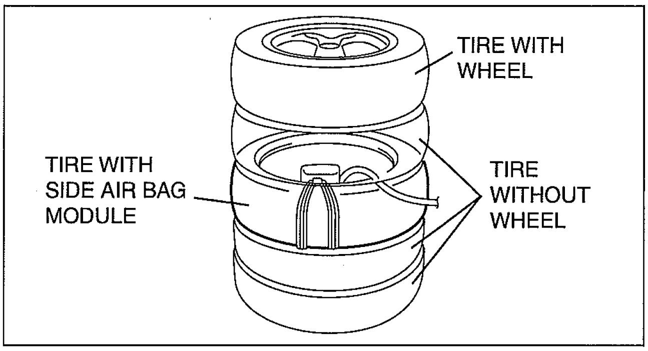
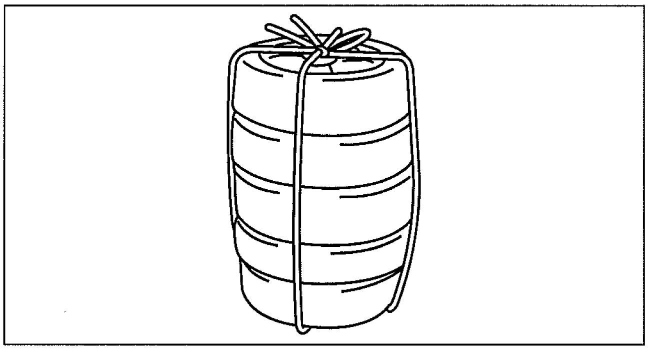
4. Stack the tire with the side air bag module on top of two tires without wheels. Stack a tire without a wheel on top of the tire with the side air bag module, and then stack another tire with a wheel on the very top.



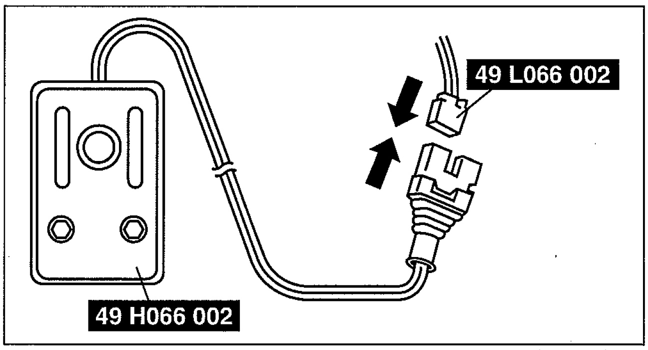
DPE810ZW1037

5. Secure the tires with wire.



A6E8130W034

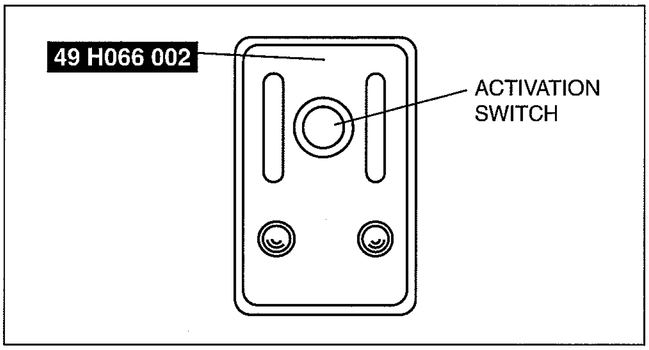
6. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
7. Connect the **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
8. Verify that the red lamp on the **SST** (Deployment tool) is illuminated.
9. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.



B3E0810W027

10. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the side air bag module.

Warning
- **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**

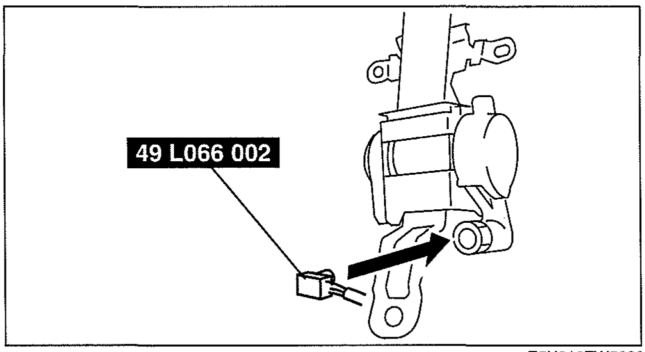


A6E8130W028

11. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).

Pre-tensioner Seat Belt

1. Remove the pre-tensioner seat belt. (See 08-11-1 SEAT BELT REMOVAL/INSTALLATION.)
2. Connect the **SST** (Adapter harness) to the pre-tensioner seat belt as shown in the figure.

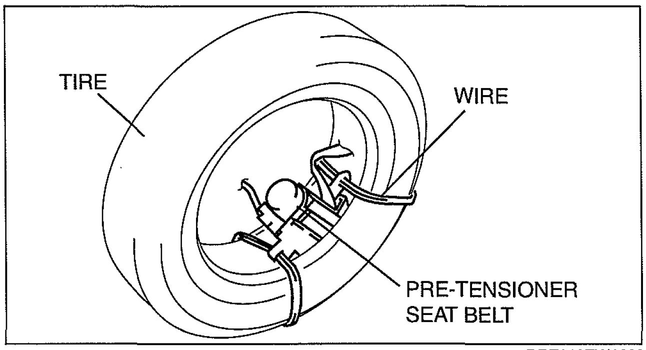


E5U810ZW5026

3. Put the pre-tensioner seat belt inside the tire and secure it to the tire by wrapping a wire (cross section of **1.25 mm^2 {0.002 in^2} or more**) through the tire and the bolt installation holes **at least 4 times.**

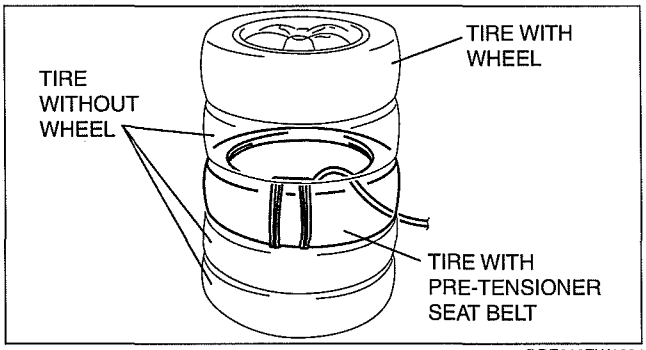
Warning

- **If the pre-tensioner seat belt is not properly secured to the tire, serious injury may occur when the pre-tensioner part is deployed. When installing the pre-tensioner seat belt to the tire, make sure the pre-tensioner part is inside the tire.**



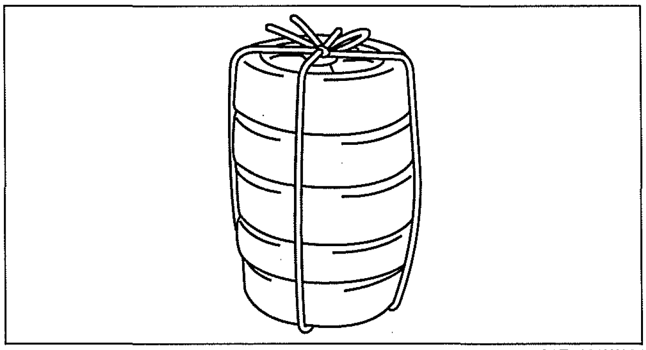
DPE810ZW1033

4. Stack the tire with the pre-tensioner seat belt on top of two tires without wheels. Stack a tire without a wheel on top of the tire with the pre-tensioner seat belt, and then stack another tire with a wheel on the very top.



DPE810ZW1034

5. Secure the tires with wire.



A6E8130W034

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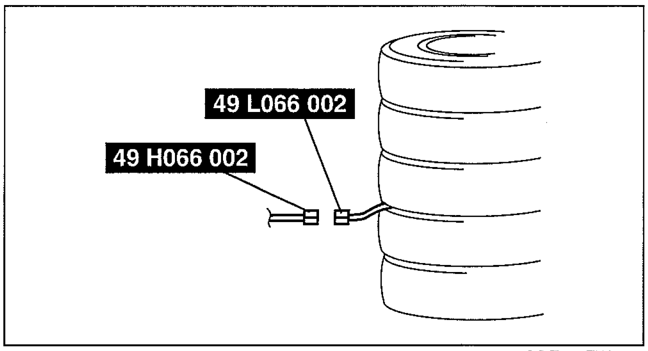
6. Connect the **SST** (Deployment tool) to the **SST** (Adapter harness).
7. Connect the **SST** (Deployment tool) to the battery. Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.
8. Verify that the red lamp on the **SST** (Deployment tool) is illuminated.
9. Verify that all persons are standing **at least 6 m {20 ft}** away from the vehicle.

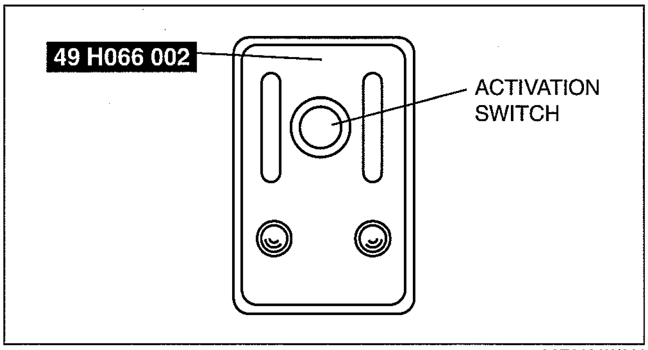


DPE810ZW1035

10. Press the activation switch on the **SST** (Deployment tool) to operate (deploy) the pre-tensioner seat belt.

Warning
- **The air bag module is very hot immediately after it is operated (deployed). You can be burned. Do not touch the air bag module for at least 15 min after deployment.**

11. Disconnect the **SST** (Deployment tool) from the **SST** (Adapter harness).



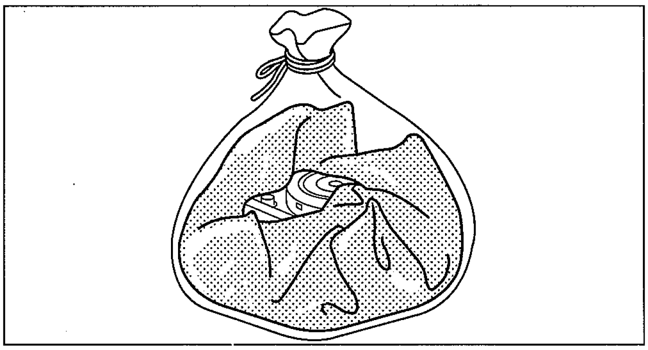
A6E8130W028

AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DISPOSAL PROCEDURES

dcf081057000w02

Warning
- **A live (undeployed) air bag module or pre-tensioner seat belt may accidentally operate (deploy) when it is disposed of and cause serious injury. Always refer to the "AIR BAG MODULE AND PRE-TENSIONER SEAT BELT DEPLOYMENT PROCEDURES" and dispose of air bag modules and pre-tensioner seat belts in a deployed condition.**
- **The air bag modules and the pre-tensioner seat belts are very hot immediately after they are deployed. You can be burned. Do not touch an air bag module and pre-tensioner seat belt for at least 15 min after deployment.**
- **Pouring water on the deployed air bag module and pre-tensioner seat belt is dangerous. The water will mix with the residual gases to form a gas that can make breathing difficult. Do not pour water on the deployed air bag module and pre-tensioner seat belt.**
- **The deployed air bag module or pre-tensioner seat belt may contain deposits of sodium hydroxide, a caustic byproduct of the gas-generated combustion. If this substance gets into your eyes or on your hands, it can cause irritation and itching. When handling the deployed air bag module and pre-tensioner seat belt, wear gloves and safety glasses.**

1. Remove the deployed air bag module or pre-tensioner seat belt.
2. Put the air bag module or pre-tensioner seat belt in a plastic bag, seal the bag, and then dispose of it.

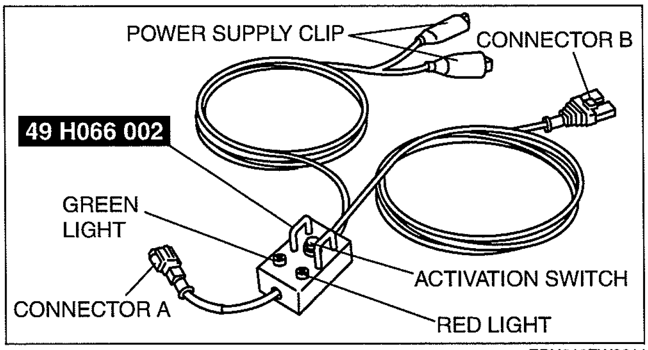


BHJ0810W023

INSPECTION OF SST (DEPLOYMENT TOOL)

dcf081001078w01

1. Before using the **SST** (49 H066 002), inspect its operation.



EPU810ZW3011

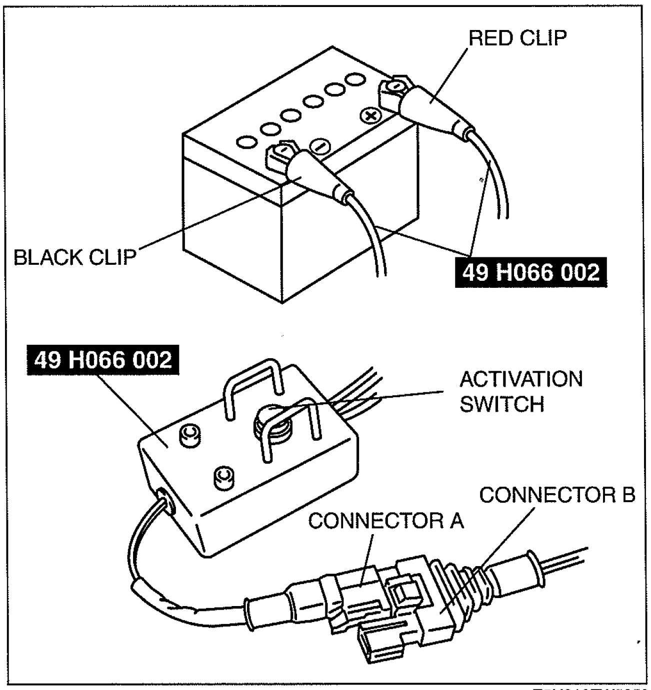
Inspection Procedure

1. Follow the steps below to inspect the **SST** (49 H066 002).
   - If not as indicated in the table, replace the **SST** (49 H066 002) because it has a malfunction.

Warning
- **Do not use a malfunctioning SST (49 H066 002), otherwise it could cause the air bag module or pre-tensioner seat belt to accidentally operate (deploy).**

Caution
- **Because the permissible voltage for the SST (49 H066 002) is 12 V, do not connect a 24 V power source because it will damage the SST. Always connect only a 12 V power source.**

Step	Inspection procedure	Light condition	
		Green	Red
1	Connect the power supply red clip to the positive battery terminal, and the black clip to the negative battery terminal.	On	Off
2	Connect connectors A and B.	Off	On
3	Press the activation switch.	On	Off

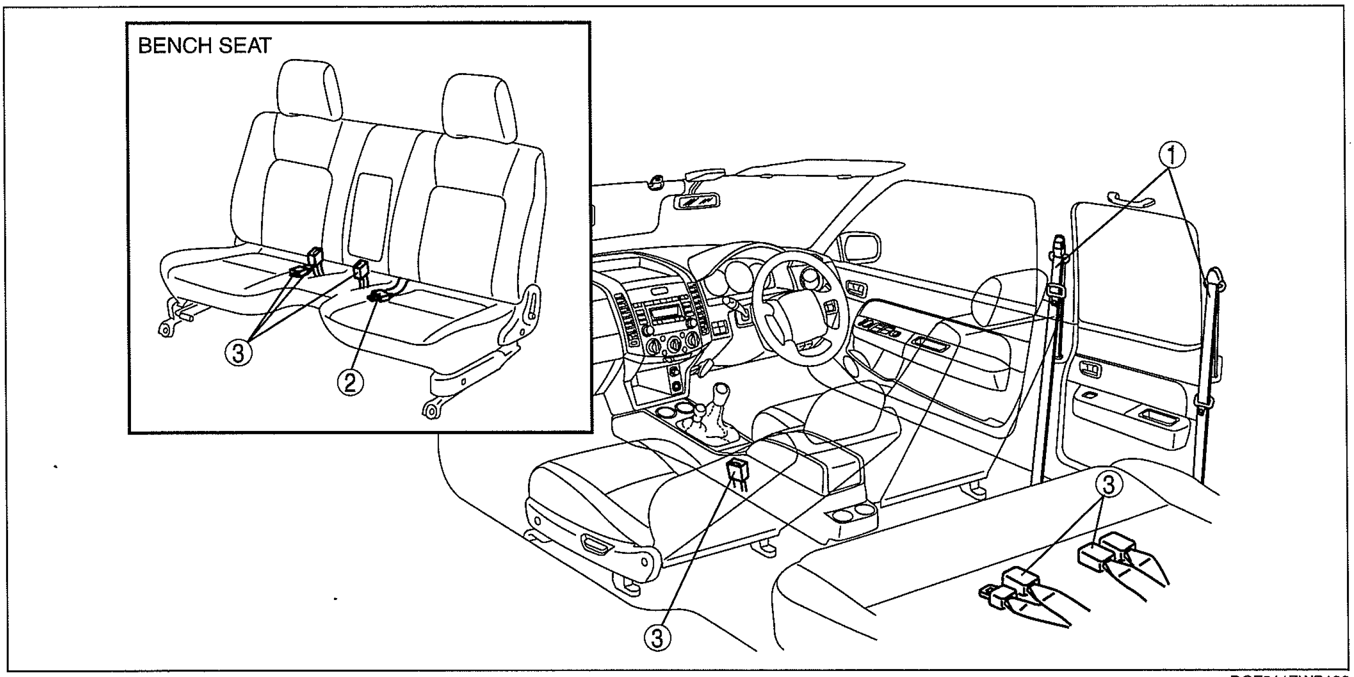


E5U810ZW5050

08-11 SEAT BELT

LOCATION INDEX

dcf081157000w01



DCF811ZWB100

1	Seat belt (See 08-11-1 SEAT BELT REMOVAL/ INSTALLATION.) (See 08-11-5 ADJUSTER ANCHOR REMOVAL/ INSTALLATION.) (See 08-11-6 SEAT BELT INSPECTION.)
2	Center seat belt (See 08-11-1 SEAT BELT REMOVAL/ INSTALLATION.)
3	Buckle (See 08-11-7 BUCKLE REMOVAL/INSTALLATION.)

SEAT BELT REMOVAL/INSTALLATION

dcf081157630w01

Warning
- **Handling the seat belt (pre-tensioner seat belt) improperly can accidentally deploy the pre-tensioner seat belt, which may seriously injure you. Read the service warnings and cautions before handling the seat belt (pre-tensioner seat belt). (See 08-10-2 SERVICE WARNINGS.) (See 08-10-3 SERVICE CAUTIONS.)**

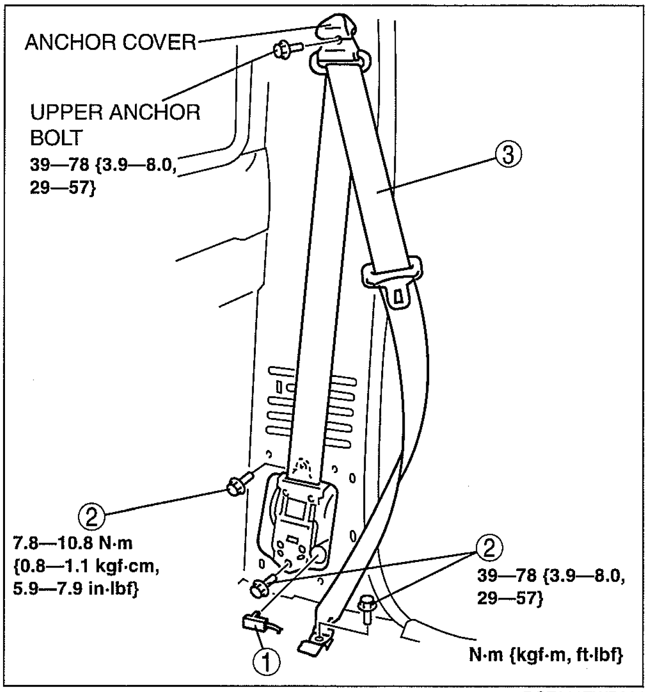
Caution
- **The ELR (emergency locking retractor) has a spring that will unwind if the retractor cover is removed. The spring cannot be rewound by hand. If this occurs, the ELR will not work properly. Therefore, do not disassemble the retractor.**

Regular Cab (Driver-side and passenger-side)

1. Turn the engine switch to the LOCK position. (vehicle with pre-tensioner seat belt)
2. Disconnect the negative battery cable and wait for **1 min or more**. (vehicle with pre-tensioner seat belt) (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
3. Remove the anchor cover.

1	Connector (vehicle with pre-tensioner seat belt) (See 08-11-2 Connector removal note.)
2	Bolt
3	Seat belt

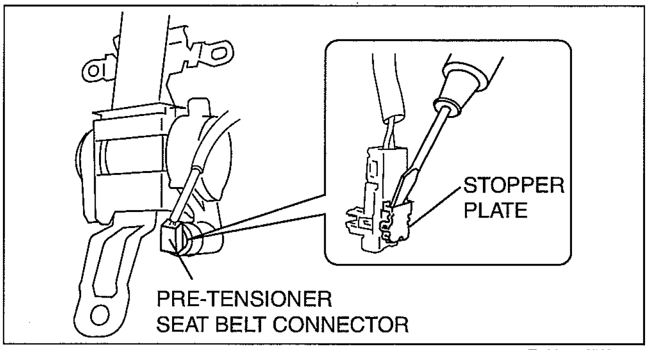
4. Remove the upper anchor bolt.
5. Remove the B-pillar trim. (See 09-17-15 B-PILLAR TRIM REMOVAL/INSTALLATION.)
6. Remove in the order indicated in the table.
7. Install in the reverse order of removal.
8. Turn the engine switch to the ON position. (vehicle with pre-tensioner seat belt)
9. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out. (vehicle with pre-tensioner seat belt)
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.



DCF811ZWB001

Connector removal note

1. Using a screwdriver, pry out the pre-tensioner seat belt connector stopper plate.



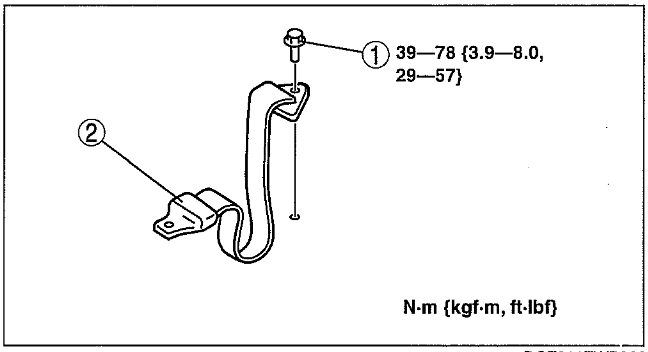
E5U811ZW5003

Regular Cab (center)

1. Remove in the order indicated in the table.

1	Bolt
2	Center seat belt

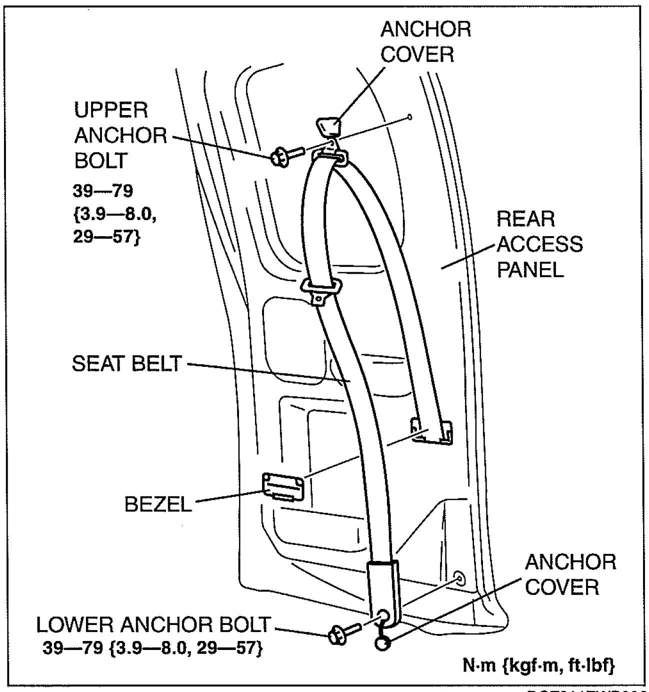
2. Install in the reverse order of removal.



DCF811ZWB002

Stretch Cab (with rear access system)

1. Turn the engine switch to the LOCK position. (vehicle with pre-tensioner seat belt)
2. Disconnect the negative battery cable and wait for **1 min or more**. (vehicle with pre-tensioner seat belt) (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
3. Remove the bezel.
4. Remove the anchor cover.
5. Remove the upper anchor bolt and lower anchor bolt.
6. Remove the Rear access panel trim. (See 09-17-18 REAR ACCESS PANEL TRIM REMOVAL/ INSTALLATION.)
7. Remove the seat belt through the seat belt hole.



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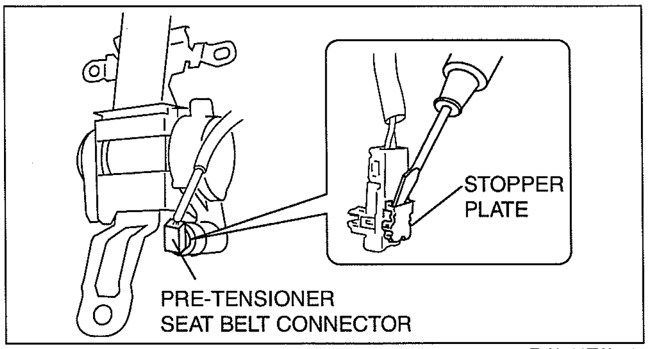
8. Using a screwdriver, pry out the pre-tensioner seat belt connector stopper plate. (vehicle with pre-tensioner seat belt)
9. Remove the bolts from the retractor.



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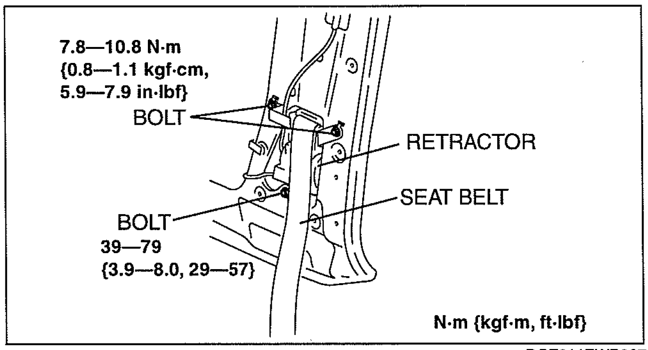
10. Remove the retractor.
11. Install in the reverse order of removal.



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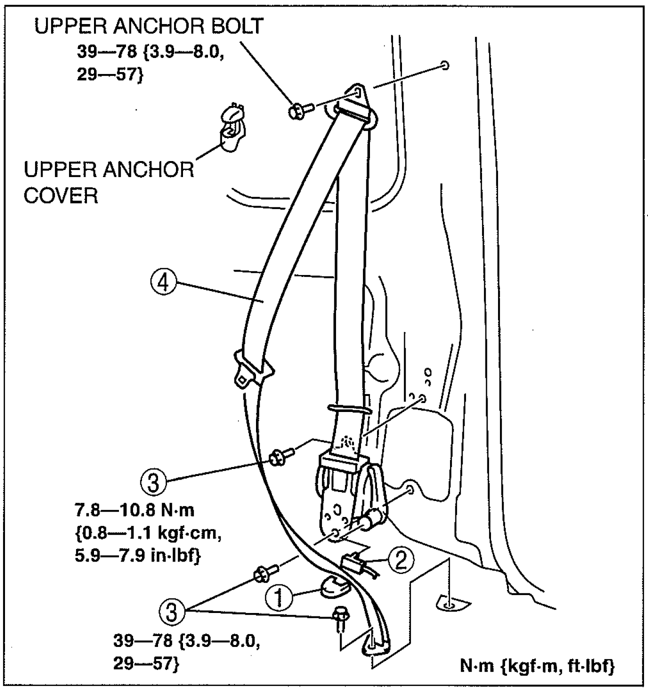
Double Cab

Front

1. Turn the engine switch to the LOCK position. (vehicle with pre-tensioner seat belt)
2. Disconnect the negative battery cable and wait for **1 min or more.** (vehicle with pre-tensioner seat belt) (See 01-17-2 BATTERY REMOVAL/INSTALLATION.)
3. Remove the following parts:
   (1) Rear scuff plate (See 09-17-21 REAR SCUFF PLATE REMOVAL/INSTALLATION.)
   (2) Rear seat back (See 09-13-5 REAR SEAT REMOVAL/INSTALLATION.)
   (3) Rear seat belt upper anchor installation bolt (See 08-11-5 Rear)
   (4) Remove the C-pillar trim (See 09-17-17 C-PILLAR TRIM REMOVAL/INSTALLATION.)
   (5) Front scuff plate (See 09-17-20 FRONT SCUFF PLATE REMOVAL/INSTALLATION.)
   (6) Remove the B-pillar lower trim (See 09-17-16 B-PILLAR LOWER TRIM REMOVAL/INSTALLATION.)
   (7) Front seat belt upper anchor cover
   (8) Front seat belt upper anchor installation bolt
   (9) Remove the B-pillar upper trim (See 09-17-16 B-PILLAR UPPER TRIM REMOVAL/INSTALLATION.)
4. Remove in the order indicated in the table.

1	Lower anchor cover
2	Connector (vehicle with pre-tensioner seat belt) (See 08-11-2 Connector removal note.)
3	Bolt
4	Seat belt



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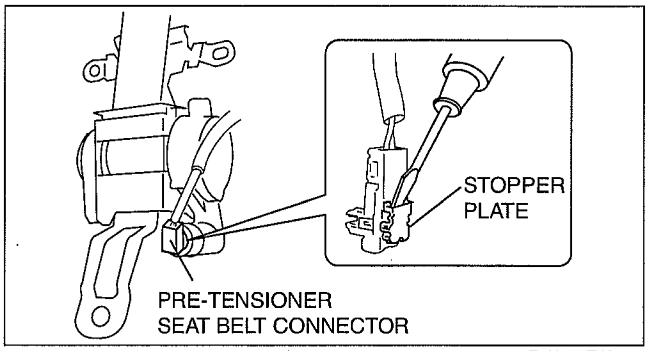
5. Install in the reverse order of removal.
6. Turn the engine switch to the ON position. (vehicle with pre-tensioner seat belt)
7. Verify that the air bag system warning light illuminates for **approx. 6 s** and goes out. (vehicle with pre-tensioner seat belt)
   - If the air bag system warning light does not operate, refer to the on-board diagnostic system (air bag system) and perform inspection of the system.

Connector removal note

1. Using a screwdriver, pry out the pre-tensioner seat belt connector stopper plate.



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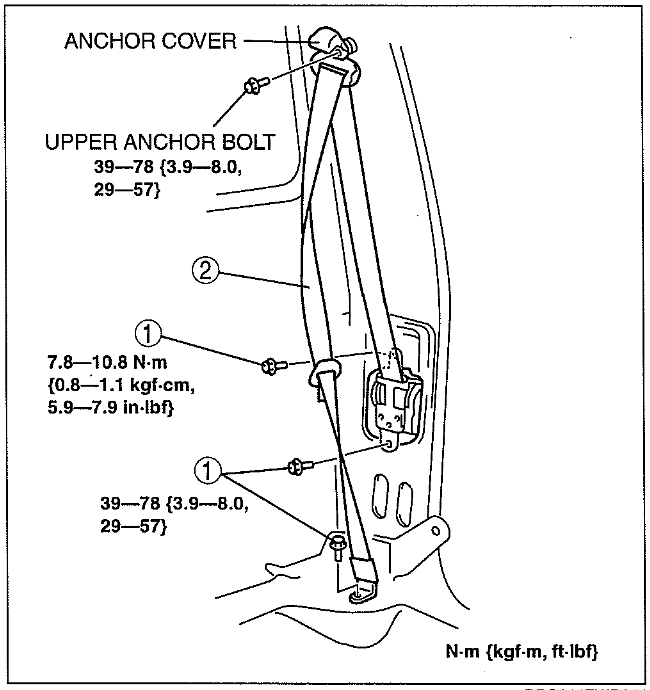
Rear

1. Remove the following parts:
   (1) Rear scuff plate (See 09-17-21 REAR SCUFF PLATE REMOVAL/INSTALLATION.)
   (2) Rear seat back (See 09-13-5 REAR SEAT REMOVAL/INSTALLATION.)
   (3) Rear seat belt upper anchor cover
   (4) Rear seat belt upper anchor installation bolt
   (5) Remove the C-pillar trim (See 09-17-17 C-PILLAR TRIM REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.

1	Bolt
2	Seat belt

3. Install in the reverse order of removal.



DBG811ZWBA01

ADJUSTER ANCHOR REMOVAL/INSTALLATION

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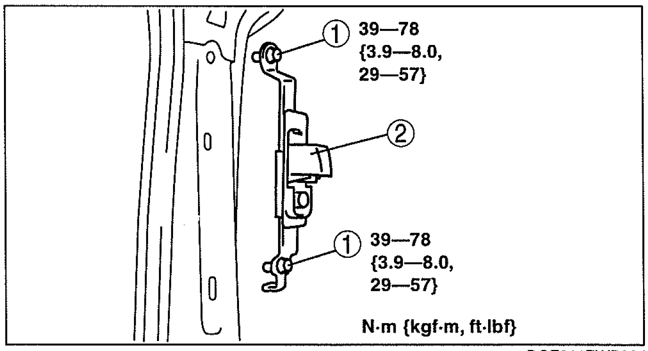
1. Remove the following parts:
   (1) Rear scuff plate (See 09-17-21 REAR SCUFF PLATE REMOVAL/INSTALLATION.)
   (2) Rear seat back (See 09-13-5 REAR SEAT REMOVAL/INSTALLATION.)
   (3) Rear seat belt upper anchor installation bolt (See 08-11-5 Rear)
   (4) Remove the C-pillar trim (See 09-17-17 C-PILLAR TRIM REMOVAL/INSTALLATION.)
   (5) Front scuff plate (See 09-17-20 FRONT SCUFF PLATE REMOVAL/INSTALLATION.)
   (6) Remove the B-pillar lower trim (See 09-17-16 B-PILLAR LOWER TRIM REMOVAL/INSTALLATION.)
   (7) Front seat belt upper anchor installation bolt (See 08-11-4 Front)
   (8) Remove the B-pillar upper trim (See 09-17-16 B-PILLAR UPPER TRIM REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.

1	Bolt
2	Adjuster anchor

3. Install in the reverse order of removal.



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SEAT BELT INSPECTION

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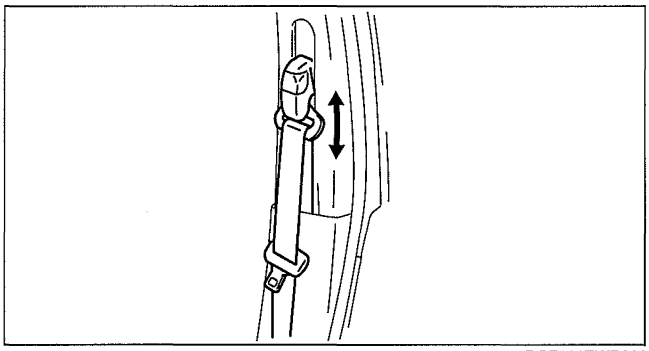
Belt

1. Verify that the belt is installed properly with no twists or kinks.
2. Verify that there is no damage to the seat belt and no deformation of the metal fittings.
   - If there is any malfunction, replace the seat belt.

ELR

1. Verify that the belt can be pulled out smoothly, and that it retracts smoothly.
   - If there is any malfunction, replace the seat belt.
2. Verify that the retractor locks when the belt is pulled quickly.
   - If there is any malfunction, replace the seat belt.
3. Remove the retractor.



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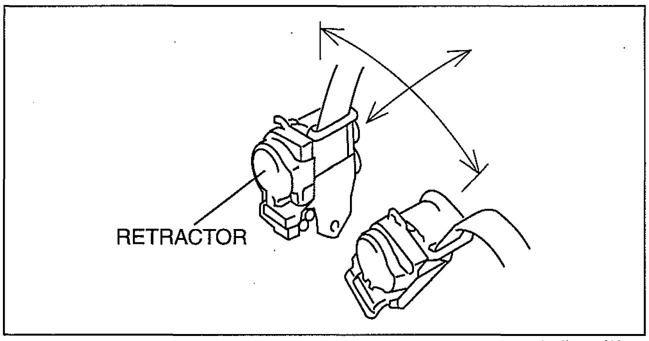
4. While pulling the seat belt out, make sure that the seat belt does not lock when the retractor is tilted slowly **up to 15°** from the mounted position and locks when the retractor is tilted **40° or more**.
   - If there is any malfunction, replace the seat belt.



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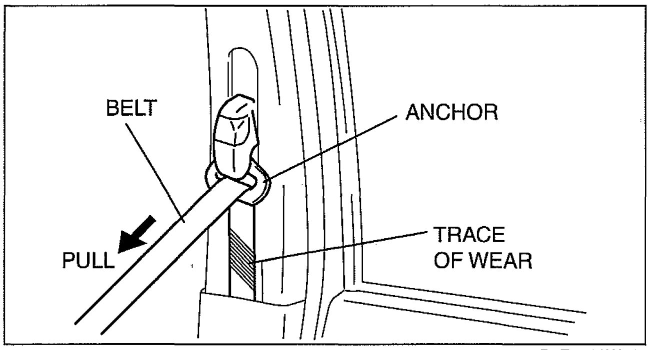
Load Limiter Retractor

Warning

- **When the load limiter operates, the belt and anchor rub against each other strongly leaving wear tracks. If the seat belt is used in this state, the seat belt will not function at its full capability and there is the possibility of serious injury to passengers. Be sure to replace the seat belt once the load limiter operates.**

1. If the vehicle has been subjected to a shock in an accident, pull the belt from the retractor and confirm that there are no wear tracks (the load limiter has not operated) by visually inspecting and feeling the belt.
   - If there is any malfunction, replace the seat belt.



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BUCKLE REMOVAL/INSTALLATION

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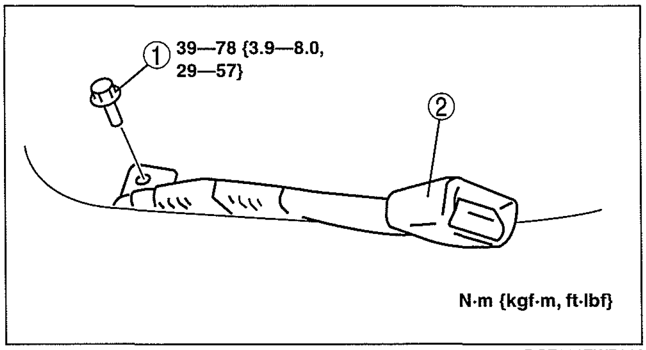
Bench Seat (Driver-side and Passenger-side)

1. Remove in the order indicated in the table.

1	Bolt
2	Buckle

2. Install in the reverse order of removal.



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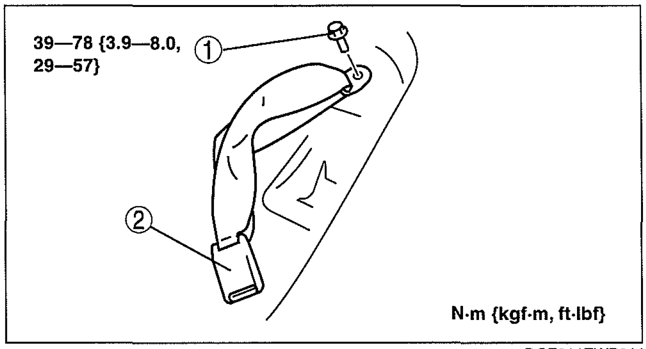
Bench Seat (Center)

1. Remove the seat. (See 09-13-2 SEAT REMOVAL/INSTALLATION.)
2. Remove in the order indicated in the table.

1	Bolt
2	Buckle

3. Install in the reverse order of removal.



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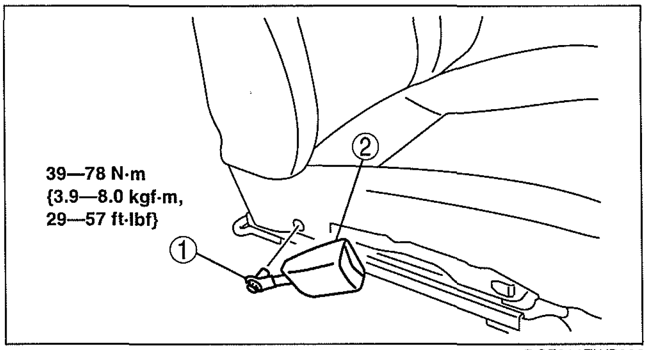
Separate Seat

1. Remove the seat or front seat. (See 09-13-2 SEAT REMOVAL/INSTALLATION.) (See 09-13-4 FRONT SEAT REMOVAL/INSTALLATION.)
2. Remove the side cover. (See 09-13-3 SEAT DISASSEMBLY/ASSEMBLY.) (See 09-13-4 FRONT SEAT DISASSEMBLY/ASSEMBLY.)
3. Remove in the order indicated in the table.

1	Bolt
2	Buckle

4. Install in the reverse order of removal.



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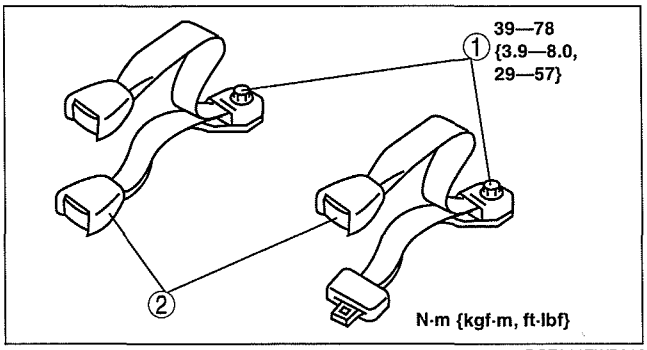
Double Cab
Rear

1. Remove in the order indicated in the table.

1	Bolt
2	Buckle

2. Install in the reverse order of removal.



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08

08-60 SERVICE TOOLS

RESTRAINTS SST

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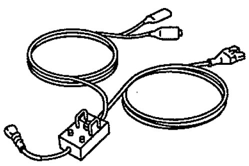
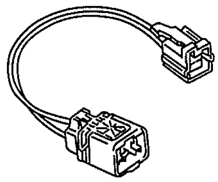
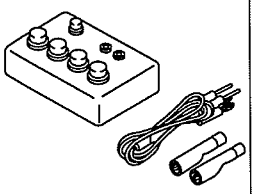
49 H066 002 Deployment tool 	49 D066 002 Adapter harness 	49 L066 002 Adapter harness 
49 N088 0A0 Fuel and Thermometer checker 	—	—